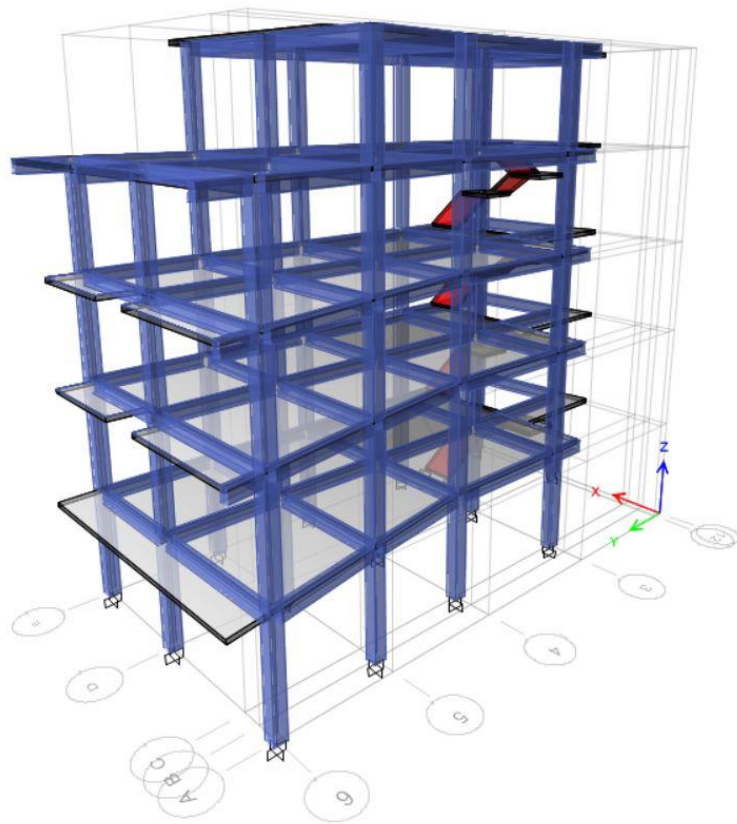


A
Detailed
Structural Analysis and Design Report
On
RCC FRAME STRUCTURE RESIDENTIAL BUILDING



Submitted to
Suryabinayak Municipality-9
Owner: Mr. Ram Pati

Prepared By
Er. Menaka Suwal

Checked By
Er. Kiran Kawan
NEC REG.: 1573 "A"

TO WHOM IT MAY CONCERN

This report comprises the summary of the structural design of the RCC frame building for Suryabinayak Municipality-9. This report consists of the design procedures adopted, the assumptions made, the inputs made in the design and the design output. During the design, it is assumed that the client will completely follow the architectural as well as the structural design. It is also assumed that the construction will be supervised by a professional engineer.

The designer will not be responsible if any alterations to the structural system is made by the client or the contractor without the prior written permission from the designer, or the alterations to the non-structural system is made such that the weight of each individual floor or the weight of the whole building is altered by more than 10% of the design weight of each floor and the total weight.

The design calculations and the derivations are limited to only a minimum to let the concerned people know the methodology adopted. However, the calculations may be provided to the client or the concerned authorities when needed, upon request.

TABLE OF CONTENTS

1.0	INTRODUCTION	1
1.1	General.....	1
1.2	Assumptions.....	1
1.3	Salient Features.....	2
1.3.1	Project Information	2
1.3.2	Building Features	2
1.3.3	Material Specification:.....	2
1.3.4	Site Condition	3
2.0	DESIGN APPROACH AND METHODOLOGY	4
2.1	General.....	4
2.2	Codes and Standard used	5
2.3	Software Used	6
3.0	ANALYSIS.....	6
3.1	Material properties	6
3.1.1	Concrete	6
3.1.2	Reinforcement Steel.....	6
3.2	Section Properties	7
3.3	Loadings.....	7
3.3.1	Load Cases	7
3.3.2	Load Combination	8
4.0	MODELLING OF THE STRUCTURE.....	9
4.1	Three Dimensional views of Model.....	9
4.2	Joint ID.....	10
4.3	Section Property.....	11
4.3.1	Beam and Columns	11
4.3.2	Slab	14
4.4	Application of loads.....	14
4.4.1	Dead Loads	15
4.4.2	Live Load	17
4.4.3	Floor Finish.....	18
4.5	Seismic Loading.....	19
4.5.1	Table 6: Mass Source for Seismic Load	19

5.0	ANALYSIS OUTPUT	20
5.1	Seismic Coefficient	20
5.2	Acceptability Criteria:	21
5.2.1	Inter-story drift:	21
5.2.2	Storey Displacement:	23
5.3	Modal time period and mass participation	24
6.0	ANALYSIS OF THE STRUCTURE	25
6.1	Axial Force Diagram	26
6.2	Shear Force Diagram	26
6.3	Bending Moment Diagram	27
6.4	Reinforcement from Etabs	28
6.4.1	Columns	28
6.4.2	Beams	33
7.0	DESIGN OF STRUCTURAL ELEMENTS	37
7.1	Design of columns	37
7.2	Design of beams	40
7.3	Design of Slab	43
7.4	Design of Staircase	47
7.5	Design of Footing	51
8.0	CONCLUSIONS AND RECOMMENDATION	60
9.0	REFERENCES	62

1.0 INTRODUCTION

1.1 General

This report summarizes the structural analysis and design of RC framed building. It has planned to utilize the building for residential purpose. The structural analysis in this building considers three floors only. The aim of design is the achievement of an acceptable probability that structures being designed will perform satisfactorily during their intended life. With an appropriate degree of safety, they should sustain all the loads and deformations of normal construction and use and have adequate durability and adequate resistance to the effects of misuse and fire. Structural Analysis of the concerned building has been done in details with analysis and designs software.

The analysis and design has been based on the prevailing codes that are in practice in Nepal, the National Building code of Nepal (NBC105:2025) and the IS code at places if required. This report consists of the design procedures adopted, the assumptions made, the inputs made in the design and the design output.

1.2 Assumptions

The following assumptions are taken into consideration in the seismic resistant analysis and design of structures:

- Adequate supervision and quality systems are provided during execution of the works.
- Construction is carried out by personnel having the appropriate skill and experience.
- Construction materials and products confirm to the pertinent codes and specifications.
- The structure is adequately maintained.
- The structure is used in accordance with the design brief.
- An earthquake is not likely to occur simultaneously with maximum flood, wind, waves or tides.
- Resonance as visualized under steady state sinusoidal excitation will not occur, as the small duration of earthquake is not enough to build up resonance amplitudes.
- Subsoil does not considerably settle or slide due to earthquake at the site of structure.

This report is divided into three chapters. Chapter one begins with introductions and general description Chapter two comprises the numerical modeling, analysis and design; Chapter three presents the summary and recommendations.

1.3 Salient Features

1.3.1 Project Information

Type of building	:	Residential Building
Location	:	Suryabinayak Municipality -9, Nangkheh , Bhaktapur
Plinth Area	:	1135.61 Square ft.
No. of Storey	:	4 & half Storey
Total Floor Area	:	5654.61 Square ft.

1.3.2 Building Features

The building has some special features which are listed below:

Type of Building	:	Special RC Moment Resisting Frame Structure
Shape	:	Rectangular Shape
Plinth level	:	As per architectural drawing
Roof floor Type	:	Accessible, Terrace
Walls	:	Brick walls
Footing Type	:	Strap Footing, Isolated Footing
Depth of foundation	:	Min'n 5 ft. below ground level or as per site condition
Dimension	:	51'-11" x 27'-9"
Storey Height	:	9'-6" (remaining) 10'-6" (Ground and 1st floor)
Total Height	:	52'-6" (from Plinth Level)
No of Storey	:	Ground + Three & half Storey

1.3.3 Material Specification:

Considering Architectural, Economic and strength demands reinforced cement concrete (RCC) is used as the major structural material. The selected material also confirms the availability and ease in construction.

Concrete Grade : M25 for all Column, Beam, Slab, Foundation etc.
Steel Grade : Fe500 for Longitudinal Reinforcement and Shear/Tie Reinforcement

Detailed Properties of Material

A. Concrete

M25 = 25 MPa

Characteristic value of cube compressive strength: Cube 150 x 150 x 150 mm (5%-quartile = no more than 5% of cubes tested at 28 days are expected to fail)

f_{ck} , cube = 25 MPa

f_{cm} = 26.6 MPa (Mean compressive strength: $f_{cm} = f_{ck} + 1.65 \times 4.0$ N/mm²)

f_{ctm} = 2.6 MPa (Mean value of concrete tensile strength)

τ_{ck} = 1.11 MPa (Characteristic value of concrete shear strength: $= 0.25 \sqrt{f_{ck}}$)

E_{cm} = 22,360.7 MPa Mean characteristic modulus of elasticity: $5000 \sqrt{f_{ck}}$

Density = 25 KN/m³

B. Reinforcement

Steel grade = Fe500

Type = Deformed bar

f_y = 500 MPa Characteristic proof strength at 0.2% yield

ϵ_{uk} $\geq$ 14.5% Ultimate Elongation

E = 200,000 MPa Elastic Modulus (Young's Modulus of Elasticity)

Density = 7850 Kg/m³

1.3.4 Site Condition

Type of soil : Type D, Soft Soil as per NBC 105:2025

Safe bearing pressure : 120 kN/m² (Assumed)

Seismic Zone Factor : 1.0 (for Kathmandu Valley) as per NBC 105:2025

2.0 DESIGN APPROACH AND METHODOLOGY

2.1 General

After completion of Architectural design, the layout of columns and beams are done without affecting the Architectural functions of building so far. Structure is modeled using finite element method. A three-dimensional beam element having 12 DOF with 6 DOFs at each node were used for modeling beams and columns in the building, while 24DOFs shell element with 6 DOFs at each node were used to model slab wall.

The structure is analyzed by the linear elastic theory to calculate internal actions produced by anticipated design loads. The analysis is carried out using state of art three dimensional structural analysis programs like ETABS 2022 V 22.7.0. The design loads considered as per the relevant codes of practice comprise dead load due to permanent structures, live load due to occupancy of the structure and seismic load due to anticipated earthquake possible at the proposed location. A number of load combinations are considered to obtain the maximum values of design stresses.

Following considerations is made during modeling, analysis and design.

- The structures are Special Reinforced Concrete Moment Resisting Frame (SMRF) type. Beams and columns are considered as the structural load resisting elements. Although non-structural components like wall plaster, infill walls, floor finishing, etc. has comes effects on structural performance, they are considered only as loading.
- For structural elements, M25 grade concrete are used.
- Centre-line dimensions are followed for modeling, analysis and design.
- Floor slabs are assumed to be rigid in their own plane. The slab action has been modeled by rigid floor diaphragms. Slabs are also considered in modeling. Slab is modeled as shell element.
- Beam and columns are modeled as frame elements.
- The main beams rest centrally on columns to avoid local eccentricity.
- Foundation is assumed to be fully rigid at the plinth level.
- The beam-column joint is not modeled in detail.
- Preliminary sizes of structural components are assumed by experience.
- Seismic loads were considered acting in the horizontal direction (along either of the two orthogonal directions of building) and not along the vertical direction, since it is not considered to be significant for design of structural members suitable load combinations as suggested by NBC 105: 2025 are used.

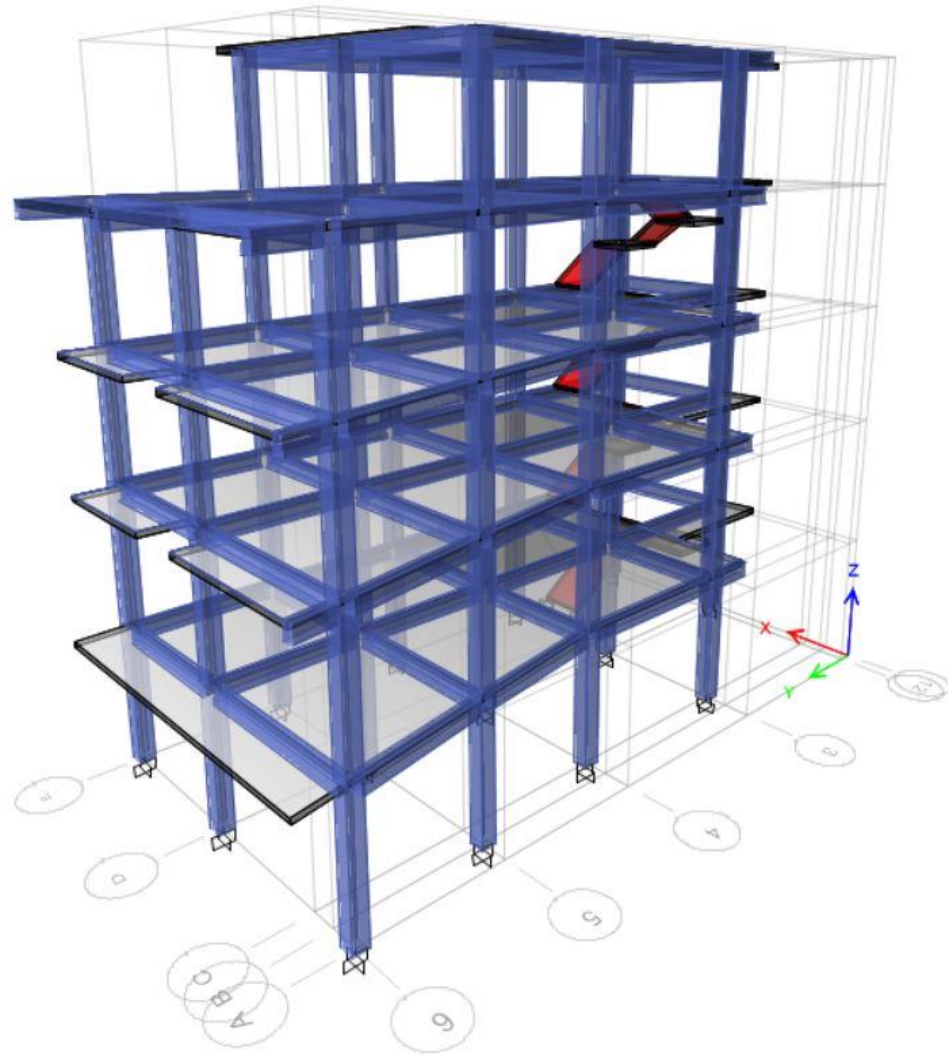


Figure 3: 3D-Model of Building

2.2 Codes and Standard used

For the structural analysis and design, the following codes and standard are followed:

- IS 456- 2000 Code of practice for plain and reinforced concrete
- IS 875-1987 Code of practice for design loads (other than earthquake) for buildings and structures
- IS 1893-2002 Criteria for Earthquake Resistant Design of Structures,
- IS 13920-1993 Code of practice for ductile detailing of reinforced concrete structures subjected to seismic forces
- NBC 105:2025 Nepal Building Code

2.3 Software Used

The following software is used for the structural analysis and design.

ETABS Ultimate C 22.7.0 For analysis and design of the structures

For structural modeling of the present building, ETABS software was used. ETABS is a special purpose finite element analysis and design program developed specifically for building systems. With ETABS, models are defined logically floor-by-floor, column-by-column, bay-by-bay and wall-by-wall and not as a stream of non-descript nodes and elements as in general purpose programs. The software has very powerful numerical methods, design procedures and international design codes, all working from a single comprehensive database. At its core, it utilizes the same analysis engine as used by SAP2000.

Among others, ETABS can do model generation, seismic and wind load generation, finite element-based linear and non-linear static and dynamic analysis, concrete frame design (column and beam) and shear wall design.

3.0 ANALYSIS

3.1 Material properties

3.1.1 Concrete

Structural elements are designed for M25 grade as specified.

$$\begin{aligned}\text{Modulus of Elasticity } [E_c] &= 5000 \sqrt{f_{ck}} \text{ N/mm}^2 (\text{Cl. 6.2.3.1, IS 456:2000}) \\ &= 25000 \text{ N/mm}^2 \text{ for M25}\end{aligned}$$

$$\text{Poisson's Ratio } [U] = 0.2$$

$$\text{Unit Weight} = 25 \text{ KN/m}^3$$

$$\text{Characteristic Strength } [f_{ck}] = 25 \text{ N/mm}^2 \text{ for M25}$$

The structural design strength is derived from the characteristic strength multiplied by a coefficient 0.67 and divided by the material partial safety factor. The partial factor for concrete in flexure and axial load is 1.5.

3.1.2 Reinforcement Steel

Characteristic strength of high yield steel is taken as 500 MPa for main longitudinal rebar and 500 Mpa for shear rebar and material partial safety factor is to be 1.15.

Modulus of Elasticity $[E_s] = 2 \times 10^5 \text{ N/mm}^2$

Poisson's Ratio $[U] = 0.3$

3.2 Section Properties

•Preliminary Size of Members

The preliminary sizes of Beam, Column, and Slab were chosen based on experience.

Beam : 355 mm x 355 mm
M25

Column : 355 mm x 355 mm
M25

Floor Slab Thickness : 125 mm M25

Staircase Slab Thickness : 125 mm M25

During the analysis, beam and column are modeled as frame elements whereas slabs are modeled as area element.

3.3 Loadings

The following considerations are made during the loading on the structural model:

- The loads distributed over the area are imposed on the area element and the loads distributed over the length are imposed on the frame elements whenever possible.
- Where such loading is not possible, equivalent conversion to different loading distribution is carried to load the model near the real case as far as possible.
- For lateral load, necessary calculations are performed to comply with the requirements of NBC 105: 2025.

3.3.1 Load Cases

The following load cases are used for the loading during analysis.

Table 1: Load Cases

Load Name	Load Type	Description	Unit	Remarks
DL	Dead	Self-wt of the structure	kN/m ²	
WL	Dead	Wall Load	kN/m	on beam
FFL	Dead	Floor Finish Load	kN/m ²	on slab

PWL	Dead	Partition Wall Load	kN/m ²	on slab
LL	Live	Imposed Load	kN/m ²	on slab
Stair Live	Live	Imposed Load	kN/m ²	on Staircase
Stair steps	Dead	Staircase steps load	kN/m ²	on Staircase
Water tank	Live	Imposed Load	kN/m ²	on terrace/roof
EQ _x	Earthquake	As per NBC 105		X+0.05Y
EQ _y	Earthquake	As per NBC 105		Y+0.05X

3.3.2 Load Combination

The following load combinations are specified as per NBC 105:2025:

Static Load Combination:

$$1.2DL + 1.5 LL$$

Seismic Load Combination:

$$DL + LL \pm EQ_x / EQ_y$$

$$0.9 DL \pm EQ_x / EQ_y$$

The following load combinations are used during analysis

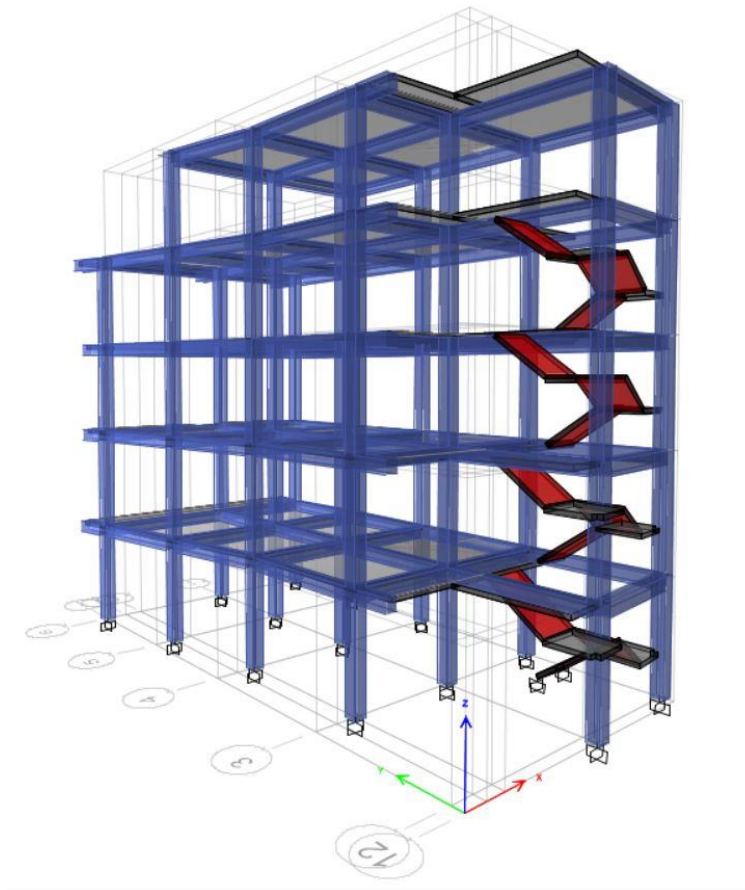
3.3.2.1 Table 2: Load Combination

S.N	Name	Type
1	1.2DL + 1.5LL	1.2DL + 1.5LL
2	DL + YLL + EQ _x	DL + YLL + EQ _x
3	DL + YLL - EQ _x	DL + YLL - EQ _x
4	DL + YLL+ EQ _y	DL + YLL+ EQ _y
5	DL + YLL-EQ _y	DL + YLL-EQ _y
6	0.9DL + EQ _x	0.9DL + EQ _x
7	0.9DL - EQ _x	0.9DL - EQ _x
8	0.9DL + EQ _y	0.9DL + EQ _y
9	0.9DL - EQ _y	0.9DL - EQ _y

4.0 MODELLING OF THE STRUCTURE

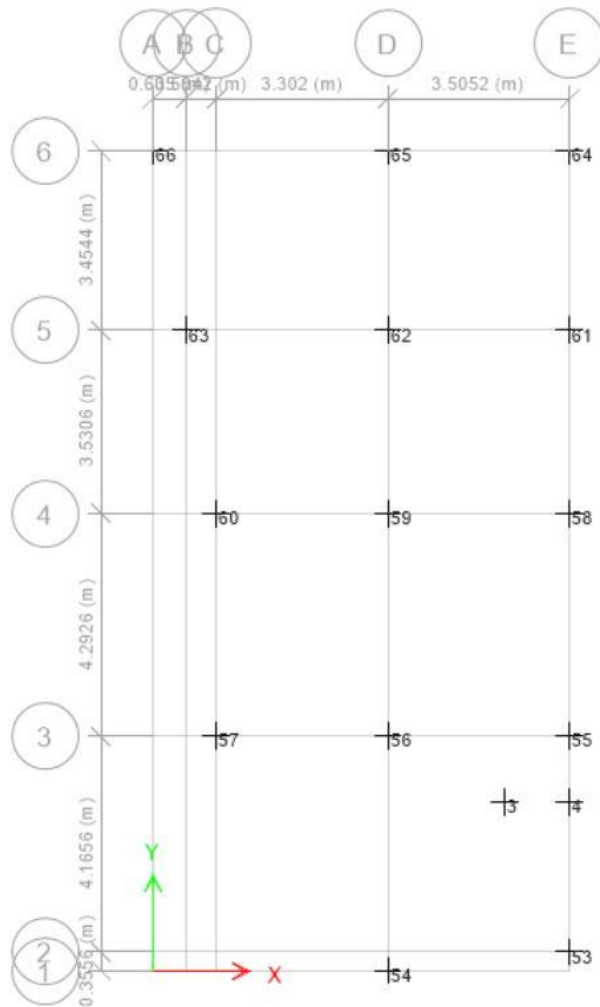
The structure is modelled in ETABS version 22, different data for modelling are shown in the following sections.

4.1 Three Dimensional views of Model



3D View

4.2 Joint ID

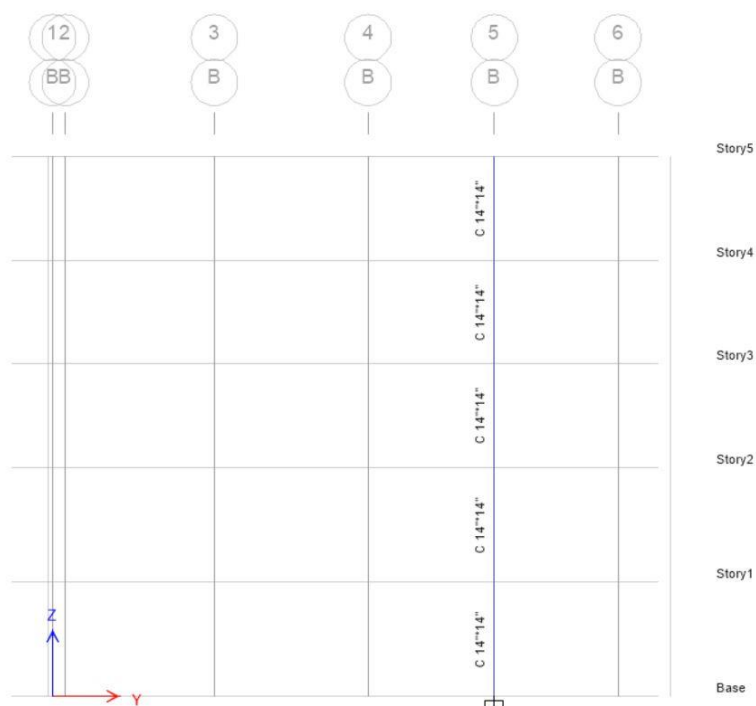


4.3 Section Property

4.3.1 Beam and Columns



Elevation View A



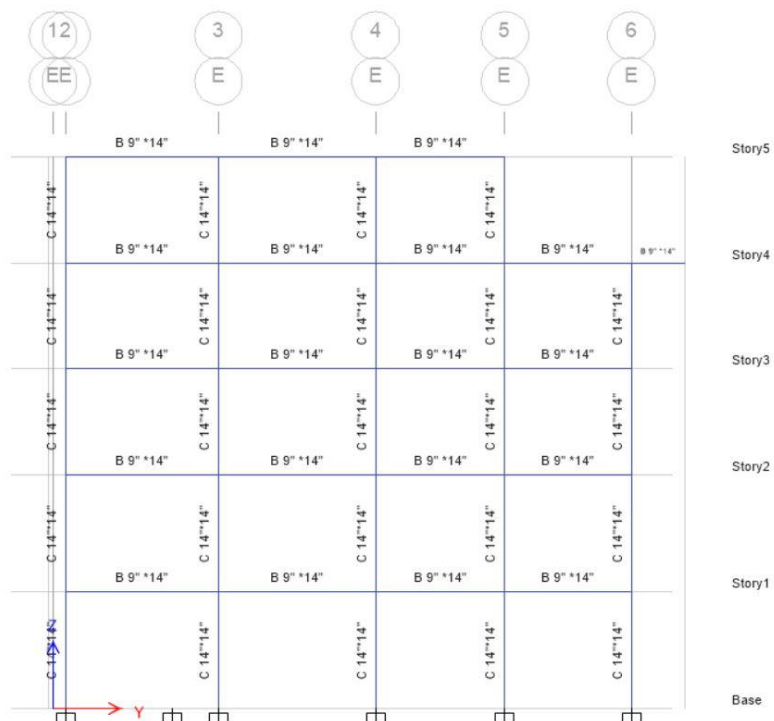
Elevation View B



Elevation View C

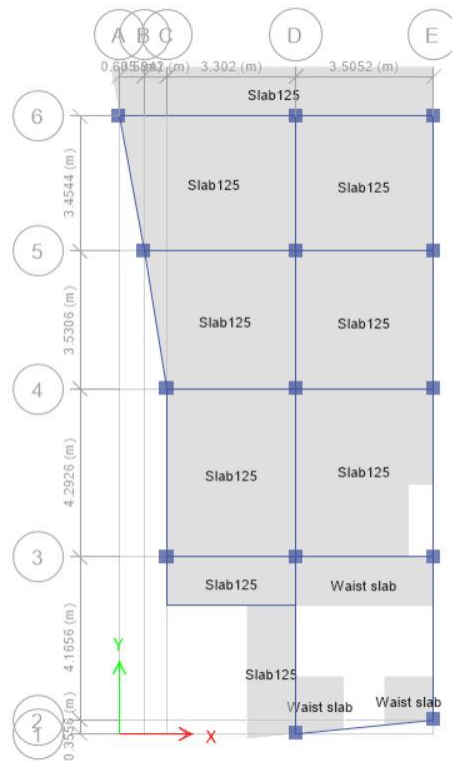


Elevation View D



Elevation View E

4.3.2 Slab



4.4 Application of loads

The loads on the building are based on Indian codes of Practices. The unit weight of different structural and non-structural elements are derived from IS 875 Part 1 and presented in Table 3. The load calculations are based on actual measured drawings. The self-weight of beams, columns and slabs are calculated by the program. Similarly the imposed loads are applied on the slab as area load in kN/m^2 and values of imposed loads are tabulated in Table 5 below.

- The weight of infill walls are calculated and applied on beams as line weight in kN/m .
- Partition wall load are assigned as uniformly distributed area load in slab as area load in kN/m^2 .
- Floor finishing load are assigned as area load in slab.
- Single type of Live load is assigned in each panel of slab.
- A frame load is applied as parapet loading on the exterior frame of the roof level.
- The roof is assumed accessible and loaded with roof live load as per Indian Standard, IS 875 -1987(part2) but this load is not considered during seismic load.

4.4.1 Dead Loads

4.4.1.1 Unit Weight (Dead Load)

Dead loads for analysis are calculated as per Indian Standard, IS 875 -1987 (part-I). Unit weights of different material used are given below.

Table 3: Unit Weight of Materials

S.N	Type	Value
1	Reinforced Concrete	25 kN/m ³
2	Brick Masonry	19.2 kN/m ³
3	Screed	22.0 kN/m ³
4	Marble	26.7 kN/m ³
5	Plaster	20.4 kN/m ³
7	Steel Rebar	78.6 KN/m ³

4.4.1.2 Dead Load Calculation

1 Unit Weights of materials

Brick masonry	19.2kN/m ³
Screed	22kN/m ³
Marble	26.7kN/m ³
Reinforced Concrete	25kN/m ³
Cement plaster	20.4kN/m ³

2 Heights of Beams, Walls & Parapet Walls

Depth of Beam	0.355m
Height of Building	3.2 at GF & FF and 2.896m remaining
Height of Parapet Wall	1.000m Wall thickness 0.115m

3 Dead Loads of Walls

- a) Dead load of 230 mm thick wall (without opening) ht. upto beam 12.56kN/m for GF & FF and 11.22KN/m for SF and above.

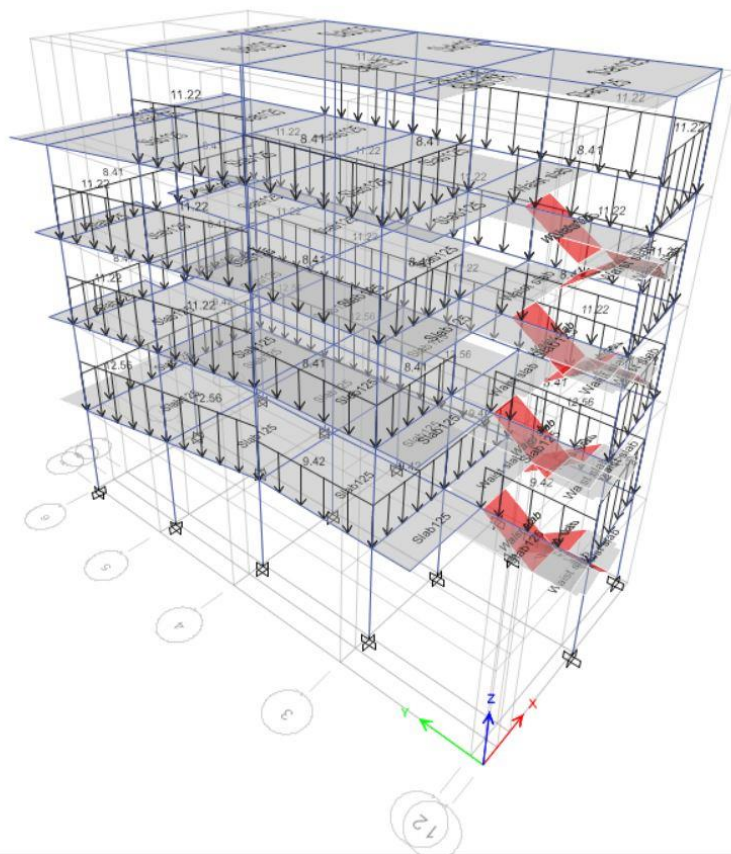
Dead load of 230 mm thick wall (without opening) ht. upto slab 13.57kN/m for GF & FF and 12.23KN/m for SF and above.

Dead load of 230 mm thick wall (with opening) ht. upto beam 9.42kN/m for GF & FF and 8.41KN/m for SF and above.

- Dead load of 230 mm thick wall (with opening) ht. upto slab 10.18kN/m for GF & FF and 9.17kN/m for SF and above.
- b) Dead load of 115 mm thick wall (without opening) ht. upto beam 6.28kN/m for GF & FF and 5.61kN/m for SF and above.
- Dead load of 115 mm thick wall (without opening) ht. upto slab 6.79kN/m for GF & FF and 6.11kN/m for SF and above.
- Dead load of 115 mm thick wall (with opening) ht. upto beam 4.71kN/m for GF & FF and 4.21kN/m for SF and above.
- Dead load of 115 mm thick wall (with opening) ht. upto slab 5.09kN/m for GF & FF and 4.58kN/m for SF and above.
- c) Dead load of parapet wall (115 mm thick) 2.02kN/m
- d)

4 Floor Loads

Thickness of slab	0.125m
Thickness of tile with plaster	0.025m
Thickness of Mosaic	0.020m
Thickness of Marble	0.020m
Thickness of Screed	0.040m
Thickness of Cement Plaster	0.0125m
Dead load of structural slab	3.125kN/m ²
Dead load of Tile	0.510kN/m ²
Dead load of Mosaic	0.462kN/m ²
Dead load of Marble	0.534kN/m ²
Dead load of screed	0.840kN/m ²
Dead load of Cement Plaster	0.255kN/m ²
Total dead load of Floor Finishes (Tile)	1.350kN/m ²
Total dead load of Floor Finishes (Mosaic)	1.302kN/m ²
Total dead load of Floor Finishes (Marble)	1.374kN/m ²
Total dead load of Floor Finishes (Cement Punning)	1.095kN/m ²
Dead load of light partition walls	1.91kN/m ²



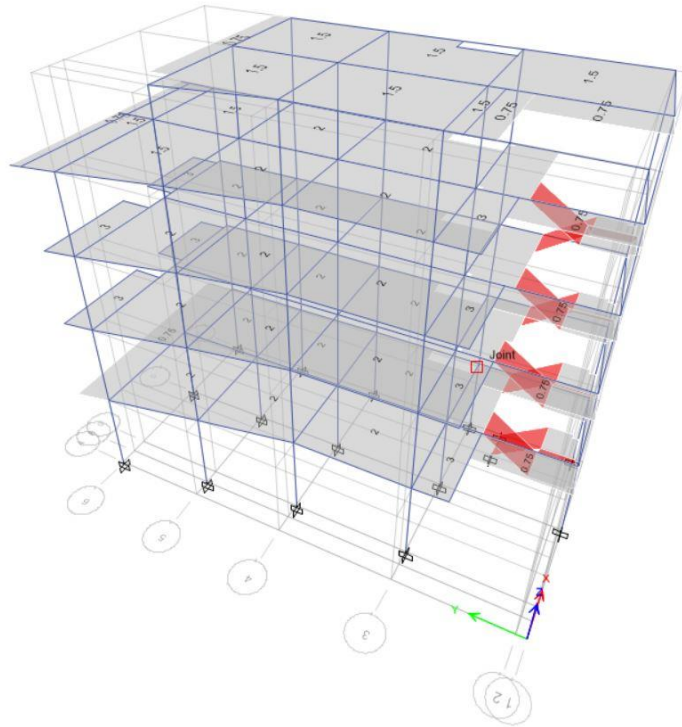
Wall load

4.4.2 Live Load

The magnitude of live load depends upon the type of occupancy of the building. These are to be chosen from code IS875:1987(part II) for various occupancies. The live load distribution varies with time. Hence each member is designed for worst combination of dead load and live loads. Live loads for residential building are given below.

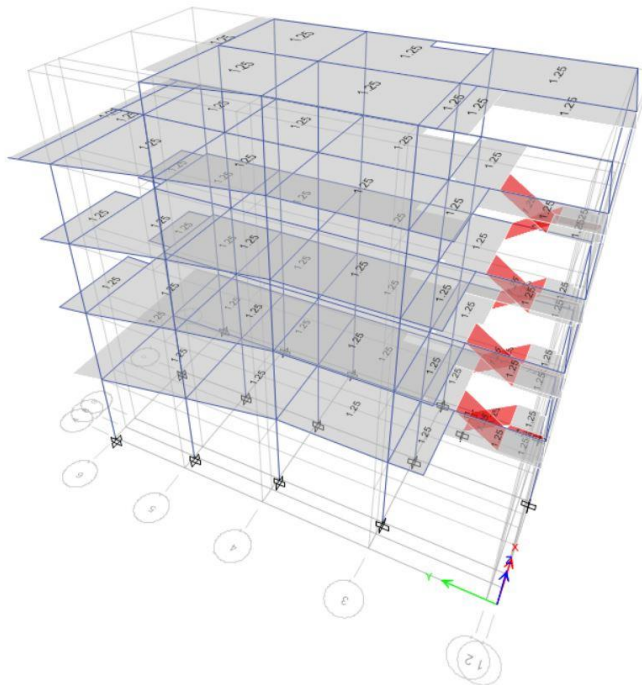
Table 5: Live Load for Residential Building

S.N	Area type	Load	Unit
1	Rooms	2.0	kN/m ²
2	Terrace (Accessible)	1.5	kN/m ²
3	Terrace (Inaccessible)	0.75	kN/m ²
4	Balcony, Staircase and Passage	3.0	kN/m ²
5	Toilet	2.0	kN/m ²



Live Load

4.4.3 Floor Finish



FF Load

4.5 Seismic Loading

The seismic load is applied to the building with user defined coefficient in ETABS 2022 calculated as per NBC 105-2025. This load case is assumed static linear and all the necessary data and calculation are given as per the following conditions.

To determine the seismic load, it is considered that the site lies in the seismic zone Gorkha according to NBC 105:2025. The soil type is considered as soft with 5% damping to determine average response acceleration. The building is analyzed as moment resisting frame with consideration of infill wall. Therefore the fundamental time period T_1 .

Detail Calculation is presented in Table 7 below.

The seismic weight is determined based on the following mass source. [Table 6, Cl.6.0.0, NBC 105: 2025]

4.5.1 Table 6: Mass Source for Seismic Load

S.N	Load Type	Scale Factor
1	Dead Load	1
2	Live Load	0.3
		0.6 for storage purpose

5.0 ANALYSIS OUTPUT

The analysis results are discussed in this chapter. Seismic coefficient method is used. The major discussion are focused on the eccentricity, story shear, inter story drift, maximum displacement and base shear along two orthogonal directions. The column and beam size and reinforcement are then designed for maximum forces.

5.1 Seismic Coefficient

The fundamental time period of the building as per NBC 105:2025

FUNDAMENTAL TIME PERIOD AS PER NBC 105: 2025

NBC 105: 2025 CL: 5.1.2 (Page 28)

$$T_1 = k_t H^{3/4} \dots\dots\dots 5.1(2)$$

Where, k_t

- = 0.075 for Moment resisting concrete frame
- = 0.085 for Moment resisting structural steel frame
- = 0.075 for Eccentrically braced structural steel frame
- = 0.05 for all other structural systems

Where,

H = Height of the building from foundation or from top of a rigid basement.

k_t	=	0.075	moment resisting concrete frame
H	=	15.0876	m
T_1	=	0.5742	seconds

The approximate fundamental time period as per
NBC 105: 2025 CL: 5.1.3 (Page 29)

T_1	=	0.7177	seconds
T_r		0.52	
T_a	=	0.5	$K = 1.108845$

BASE SHEAR COEFFICIENT AS PER NBC 105: 2025

Z	=	0.35	Kathmandu	Table 4-5
I	=	1	Residential	Table 4-6
				Clause:
$C_h(T)$	=	2.25		4.1.2
T	=	0.7177	Sec	($T_a < T < T_c$)
Soil Type	=	D		Table 4-4
α	=	2.25	check table 4.1	Table 4-1
T_a	=	0.5		Table 4-1
T_c	=	2		Table 4-1
K	=	0.8		Table 4-1
$C(T) = C_h(T) Z$				Clause:
For Ultimate Limit State	I	=	0.7875	4.1.1
For Serviceability Limit State	$C_s(T) = 0.2 C(T)$	=	0.1575	Clause: 4.2

Horizontal Base Shear Coefficient for Ultimate Limit State (Clause: 6.1.1)

$$C_d(T_1) = \frac{C(T_1)}{R_\mu \times \Omega_u}$$

$$\begin{aligned} R_{\mu} &= 4 \\ \Omega_u &= 1.5 \\ C_d(T_1) &= 0.1313 \end{aligned}$$

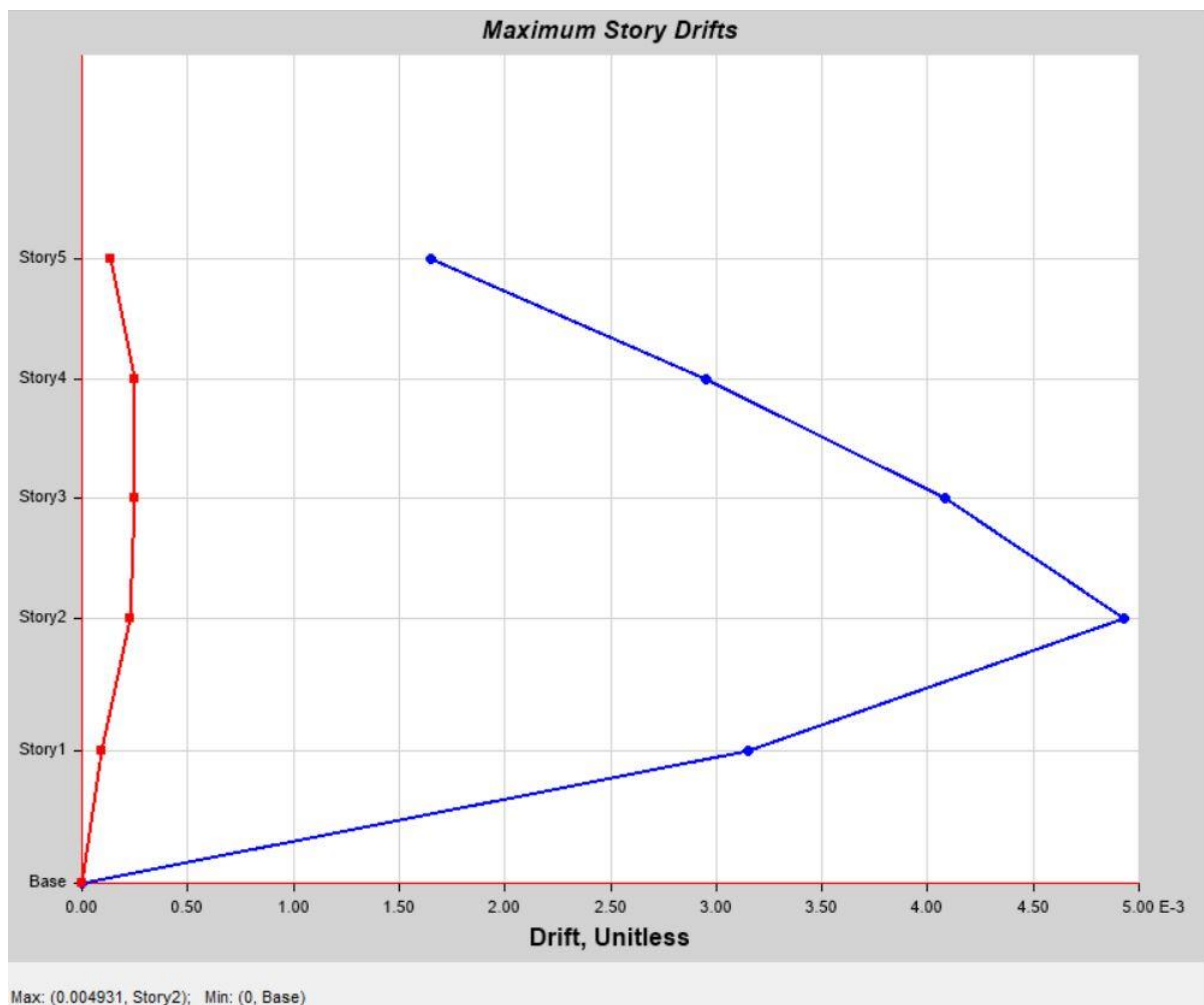
Horizontal Base Shear Coefficient for Serviceability Limit State (Clause: 6.1.2)

$$C_d(T_1) = \frac{C_s(T_1)}{\Omega_s}$$

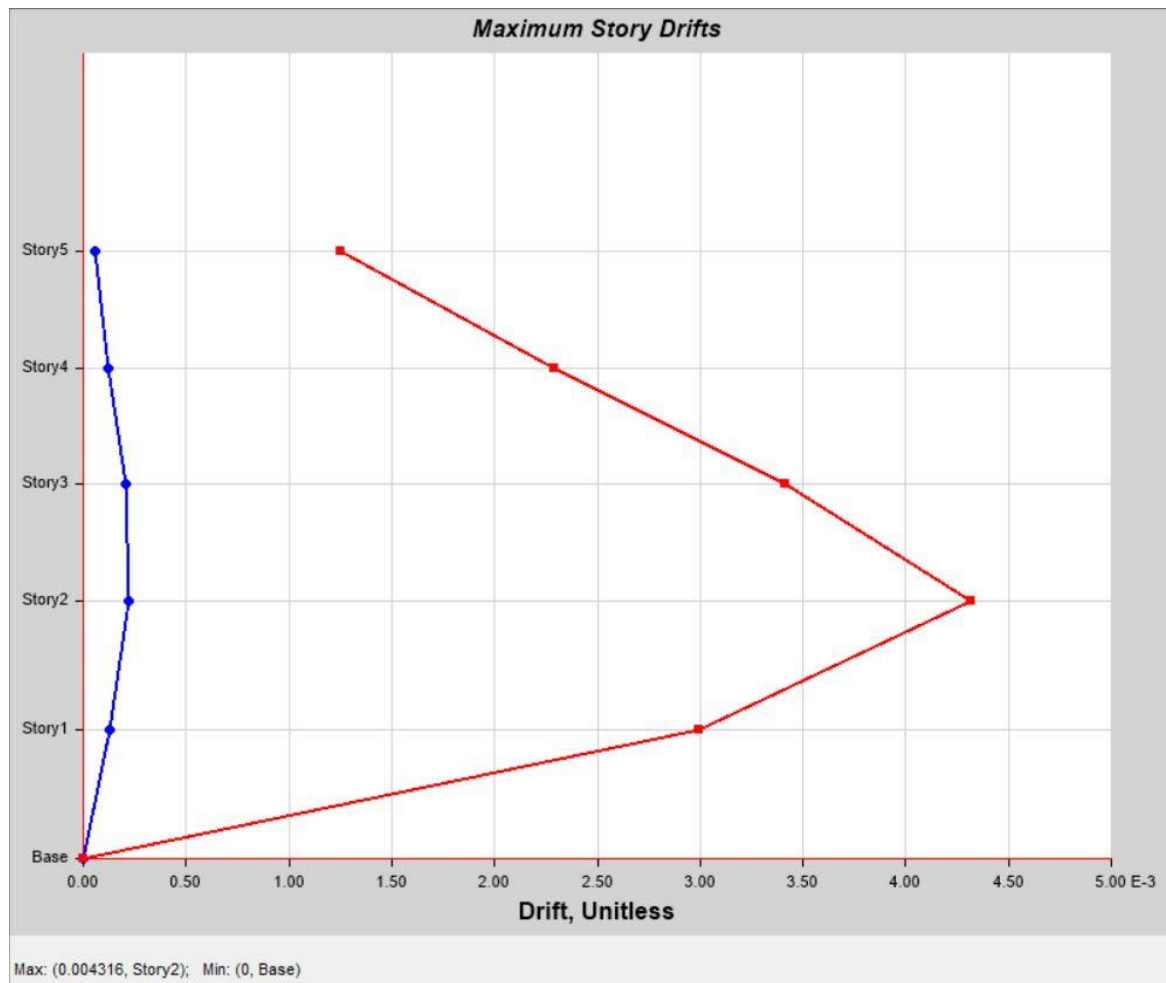
$$\begin{aligned} \Omega_s &= 1.25 \\ C_d(T_1) &= 0.1260 \end{aligned}$$

5.2 Acceptability Criteria:

5.2.1 Inter-story drift:



Inter storey drift (Load Case EQx)



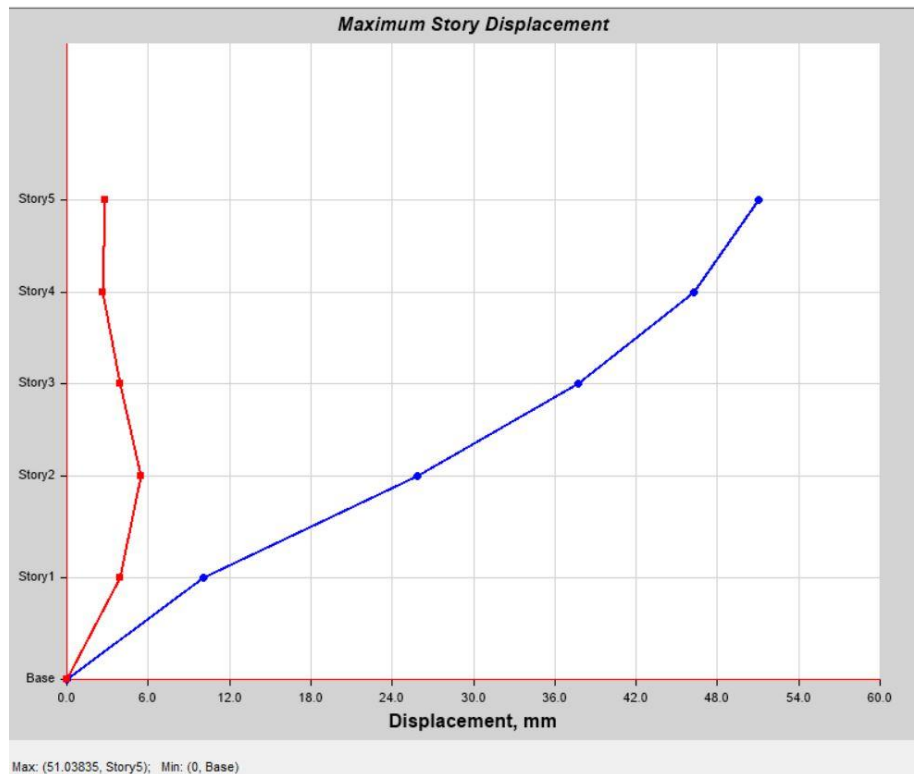
Inter storey drift (Load Case EQy)

As per cl. no. 5.5.3 of NBC 105:2025, The ratio of the inter-story deflection to the corresponding story height shall not exceed:

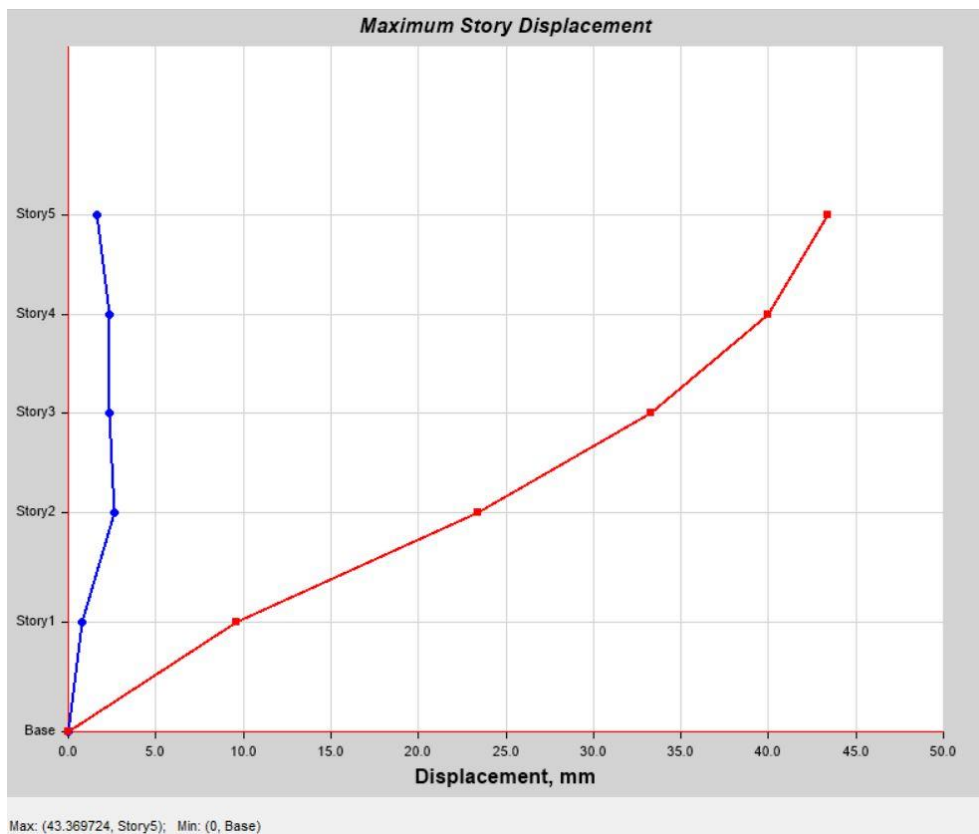
0.025 at ultimate limit state

0.006 at serviceability limit state

5.2.2 Storey Displacement:



Storey Displacement: EQx



Story displacement: EQy

5.3 Modal time period and mass participation

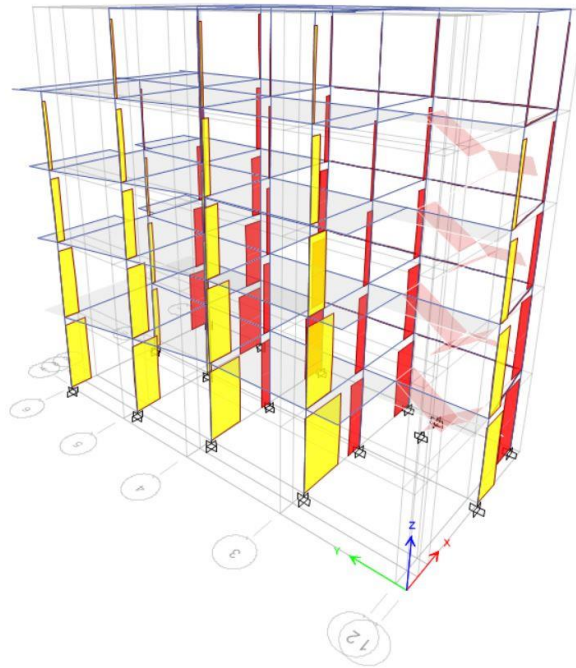
NBC 105:1994 clause 11.2 states that a sufficient number of modes shall be considered to ensure that at least 90 percent of the mass is participating in the direction under consideration. Analysis was carried out for first 10 modes so that the mass participation satisfies this criterion in both orthogonal directions. Table shows time period and mass participation ratio for all modes.

TABLE: Modal Participating Mass Ratios														
Case	Mode	Period sec	UX	UY	UZ	SumUX	SumUY	SumUZ	RX	RY	RZ	SumRX	SumRY	SumRZ
Modal	1	1.131	0.6939	0.0002	0	0.6939	0.0002	0	1.741E-05	0.179	0.1385	1.741E-05	0.179	0.1385
Modal	2	1.095	0.0048	0.8278	0	0.6987	0.828	0	0.1677	0.0003	0.0152	0.1677	0.1793	0.1537
Modal	3	1.01	0.1451	0.019	0	0.8438	0.8469	0	0.0043	0.0059	0.6872	0.172	0.1852	0.8409
Modal	4	0.344	0.0793	4.289E-05	0	0.9231	0.847	0	0.0002	0.5475	0.017	0.1722	0.7327	0.8579
Modal	5	0.34	0.0001	0.0894	0	0.9232	0.9364	0	0.6505	0.0005	0.0009	0.8227	0.7332	0.8589
Modal	6	0.31	0.0136	0.0011	0	0.9368	0.9375	0	0.0084	0.0992	0.0764	0.8311	0.8324	0.9353
Modal	7	0.194	0.0011	0.0389	0	0.938	0.9764	0	0.0805	0.0023	0.0012	0.9116	0.8346	0.9364
Modal	8	0.191	0.041	0.0011	0	0.9789	0.9775	0	0.0022	0.0829	0.0012	0.9137	0.9175	0.9376
Modal	9	0.183	9E-07	0.0018	0	0.9789	0.9793	0	0.0038	0	0.0406	0.9175	0.9175	0.9782
Modal	10	0.139	0.0094	0.0033	0	0.9883	0.9826	0	0.0137	0.0371	0.006	0.9312	0.9546	0.9841
Modal	11	0.136	0.0039	0.014	0	0.9922	0.9966	0	0.0586	0.0158	0.0001	0.9898	0.9704	0.9842
Modal	12	0.128	0.005	0.0006	0	0.9972	0.9973	0	0.0028	0.0216	0.0125	0.9926	0.992	0.9967

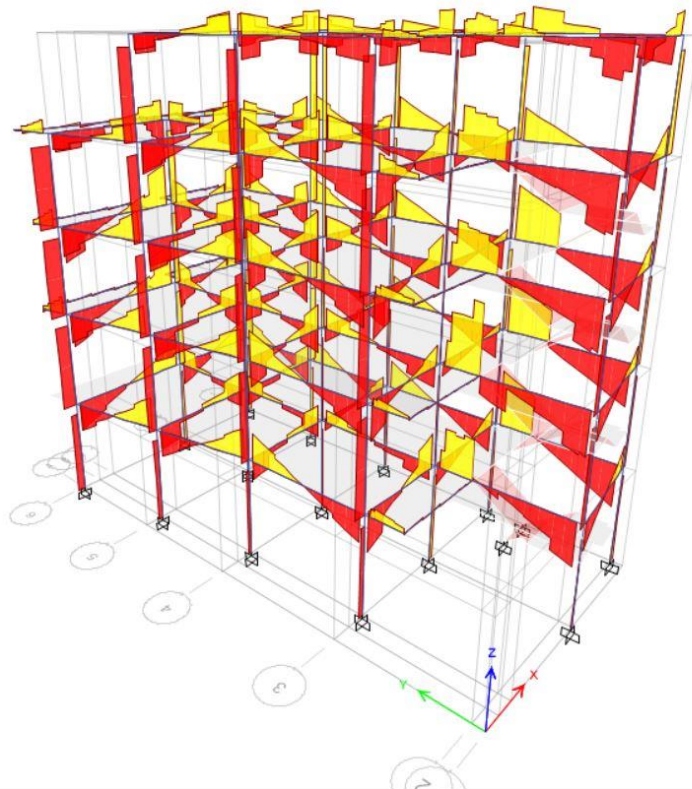
TABLE: Modal Load Participation Ratios				
Case	ItemType	Item	Static	Dynamic
			%	%
Modal	Acceleration	UX	100	99.72
Modal	Acceleration	UY	100	99.73
Modal	Acceleration	UZ	0	0



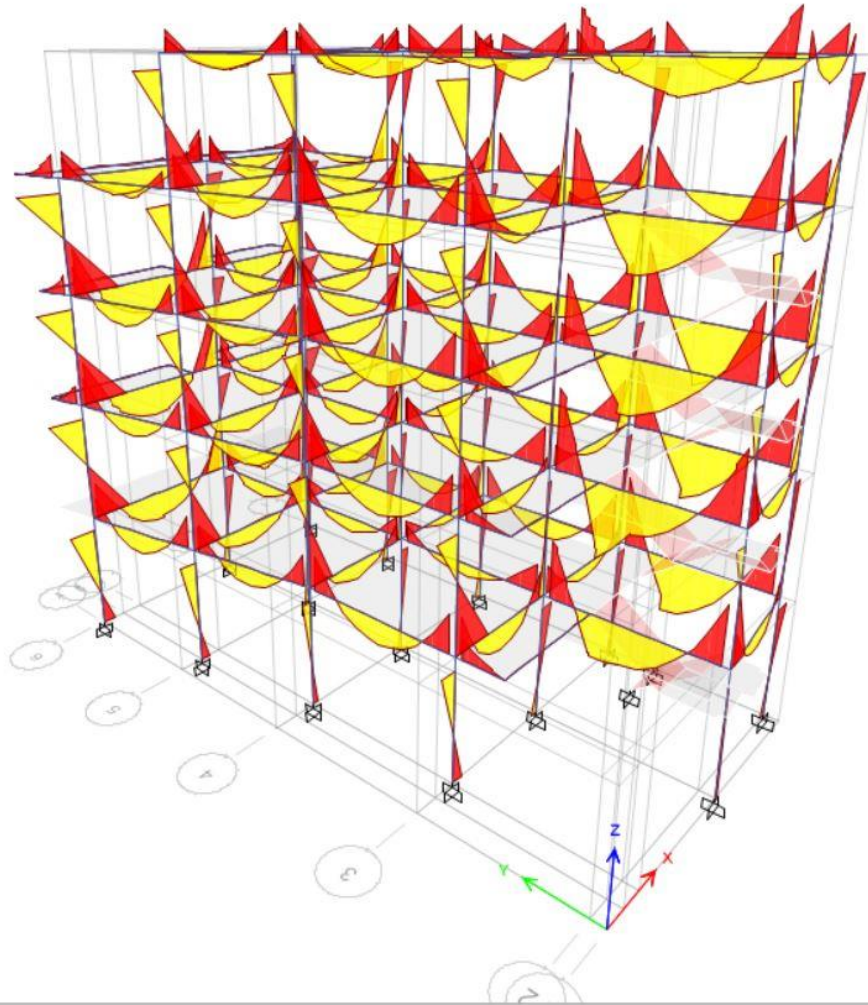
6.1 Axial Force Diagram



6.2 Shear Force Diagram



6.3 Bending Moment Diagram



6.4 Reinforcement from Etabs

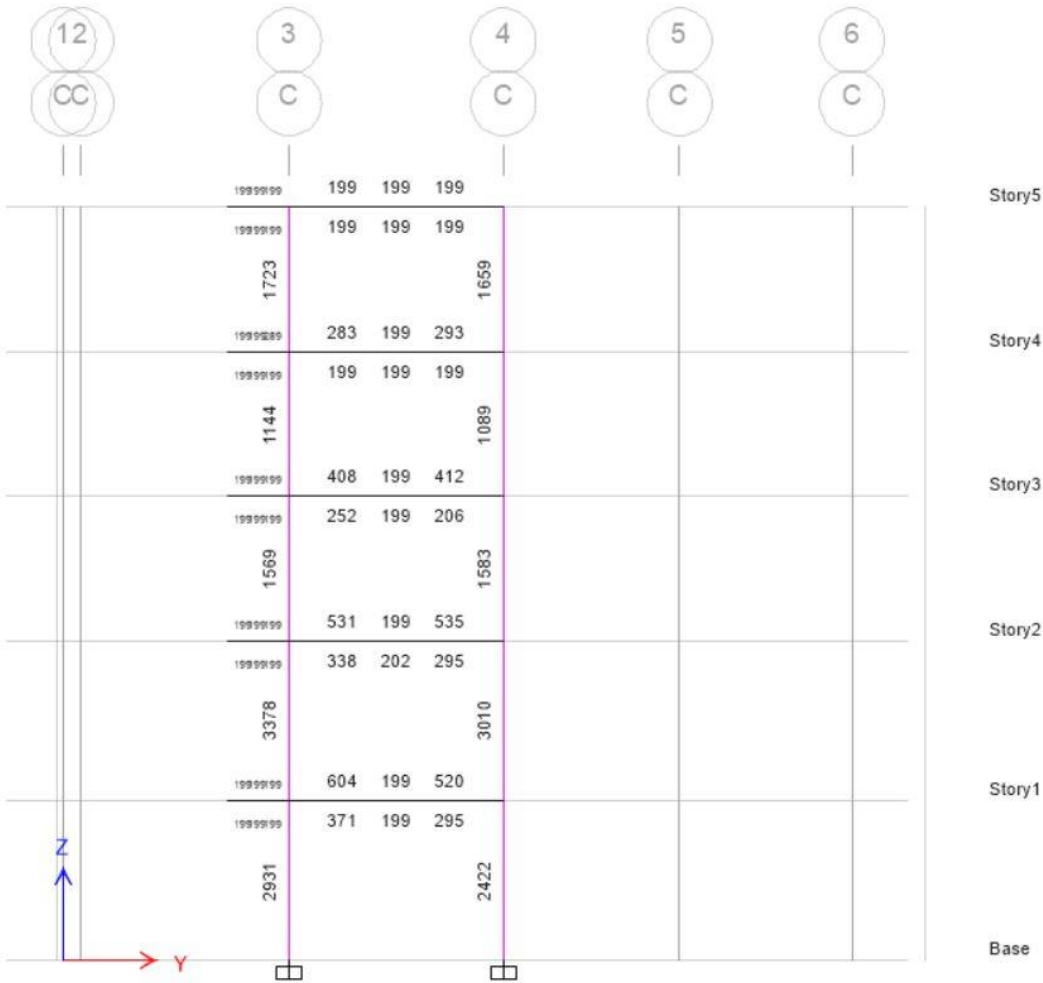
6.4.1 Columns



Column Along Grid A-A



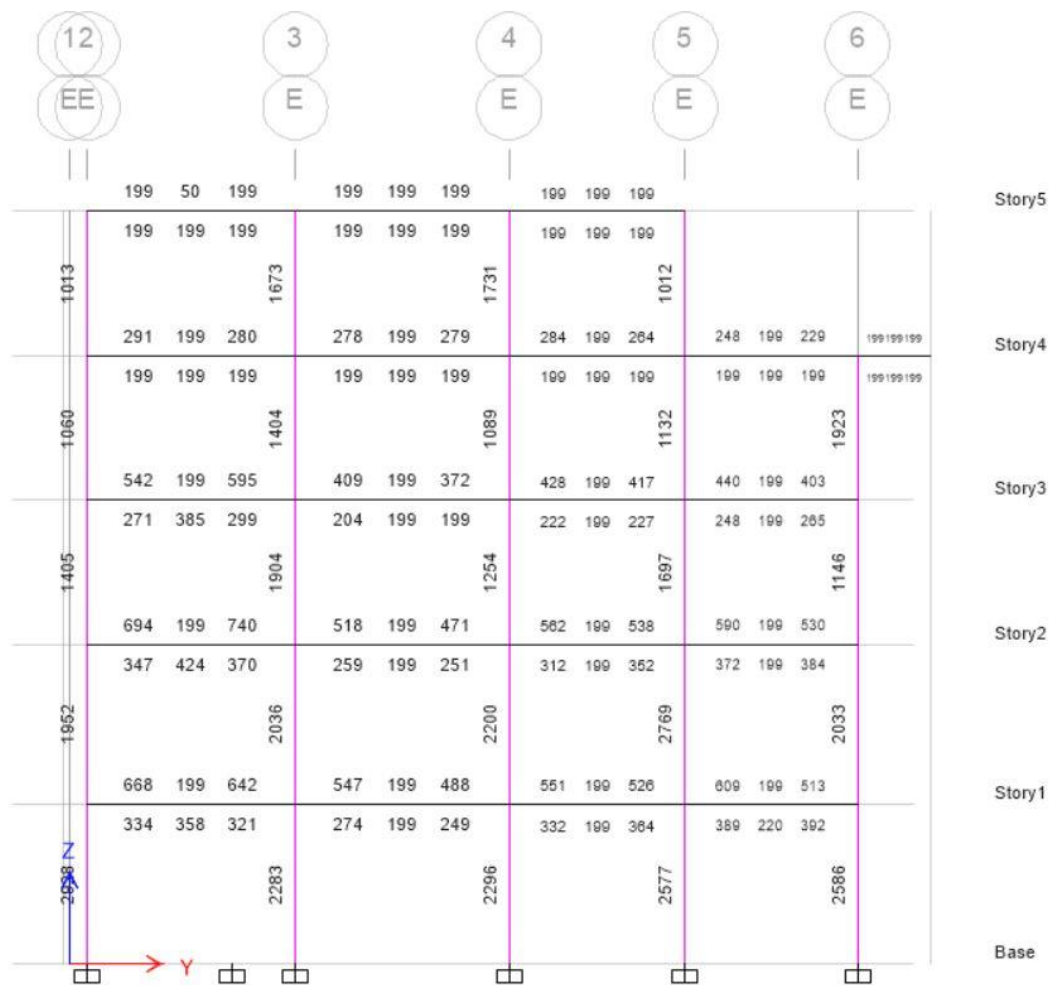
Column Along Grid B-B



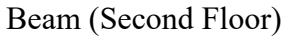
Column Along Grid C-C



Column Along Grid D-D

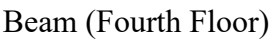


Column Along Grid E-E





Beam (Third Floor)





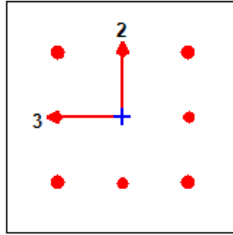
Beam (Roof Floor)

7.0 DESIGN OF STRUCTURAL ELEMENTS

7.1 Design of columns

ETABS Concrete Frame Design

IS 456:2000 + IS 13920:2016 Column Section Design (Summary)



Column Element Details

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (mm)	LLRF	Type
Story2	C17	145	C 14**14"	DL +YLL-EQxULS	0	3200.4	1	Ductile Frame

Section Properties

b (mm)	h (mm)	dc (mm)	Cover (Torsion) (mm)
355.6	355.6	78	50

Material Properties

E_c (MPa)	f_{ck} (MPa)	Lt.Wt Factor (Unitless)	f_y (MPa)	f_{ys} (MPa)
25000	25	1	500	500

Design Code Parameters

γ_c	γ_s
1.5	1.15

Axial Force and Biaxial Moment Design For P_u , M_{u2} , M_{u3}

Design P_u kN	Design M_{u2} kN-m	Design M_{u3} kN-m	Minimum M_2 kN-m	Minimum M_3 kN-m	Rebar Area mm ²	Rebar % %
517.1834	10.3437	-153.132	10.3437	10.3437	3378	2.67

Axial Force and Biaxial Moment Factors

	K Factor Unitless	Length mm	Initial Moment kN-m	Additional Moment kN-m	Minimum Moment kN-m
Major Bend(M3)	2.866043	2844.8	-42.9574	45.7386	10.3437
Minor Bend(M2)	0.921148	2844.8	1.4251	0	10.3437

Axial Compression Check for Seismic Design per IS 13920:2016 Section 7.1

Limit Compression Factor Unitless	Limit Compression Strength kN	Compression D/C Ratio Unitless	Status
0.4	1264.5136	0.408998	OK

Shear Design for V_{u2} , V_{u3}

	Shear V_u kN	Shear V_c kN	Shear V_s kN	Shear V_p kN	Rebar A_{sv} /s mm ² /m
Major, V_{u2}	65.1121	113.8675	39.4853	50.2787	394.16
Minor, V_{u3}	44.9928	113.0202	39.4853	44.9928	394.16

Joint Shear Check/Design

	Joint Shear Force kN	Shear V_{top} kN	Shear $V_{u,Tot}$ kN	Shear V_c kN	Joint Area cm ²	Shear Ratio Unitless
Major Shear, V_{u2}	N/N	N/N	N/N	N/N	N/N	N/N
Minor Shear, V_{u3}	N/N	N/N	N/N	N/N	N/N	N/N

(1.4) Beam/Column Capacity Ratio

Major Ratio	Minor Ratio
N/N	N/N

Additional Moment Reduction Factor k (IS 39.7.1.1)

A_g cm ²	A_{sc} cm ²	P_{uz} kN	P_b kN	P_u kN	k Unitless
1264.5	33.8	2689.3087	393.5624	517.1834	0.946152

Additional Moment (IS 39.7.1)

	Consider M_a	Length Factor	Section Depth (mm)	KL/Depth Ratio	KL/Depth Limit	KL/Depth Exceeded	M_a Moment (kN-m)
Major Bending (M_3)	Yes	0.889	355.6	22.928	12	Yes	48.3417
Minor Bending (M_2)	Yes	0.889	355.6	7.369	12	No	0

Notes:

N/A: Not Applicable

N/C: Not Calculated

N/N: Not Needed

Table 1: Summary of Design of Column

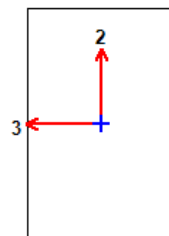
S.N.	Grid Location	Column Size	Ground Floor	First Floor	Second Floor	Third Floor	Top Floor	Stirrups
1	A6	350 x 350	8-20Ø	8-20Ø	4-20Ø 4-16Ø	4-20Ø 4-16Ø		
2	B5	350 x 350	4-25Ø 4-20Ø	4-25Ø 4-20Ø	8-20Ø	4-20Ø 4-16Ø	8-16Ø	mm C/C (at top and bottom) & @ 150 mm C/C (at mid height) in all columns
3	C3	350 x 350	8-25Ø	8-25Ø	4-25Ø 4-20Ø	8-20Ø	4-20Ø 4-16Ø	
4	C4	350 x 350	4-25Ø 4-20Ø	4-25Ø 4-20Ø	8-20Ø	4-20Ø 4-16Ø	4-20Ø 4-16Ø	
5	D6	350 x 350	8-25Ø	8-25Ø	4-25Ø 4-20Ø	8-20Ø		
6	D5	350 x 350	4-25Ø 4-20Ø	4-25Ø 4-20Ø	8-20Ø	4-20Ø 4-16Ø	4-20Ø 4-16Ø	
7	D4	350 x 350	4-25Ø 4-20Ø	4-25Ø 4-20Ø	8-20Ø	4-20Ø 4-16Ø	4-20Ø 4-16Ø	

8	D3	350 x 350	4-25Ø 4-20Ø	4- 25Ø 4- 20Ø	8-20Ø	4- 20Ø 4- 16Ø	4-20Ø 4- 16Ø	
9	D2	350 x 350	4-25Ø 4-20Ø	4- 25Ø 4- 20Ø	8-20Ø	4- 20Ø 4- 16Ø	4-20Ø 4- 16Ø	
10	E6	350 x 350	4-25Ø 4-20Ø	8- 20Ø	4-20Ø 4-16Ø	4- 20Ø 4- 16Ø		
11	E5	350 x 350	4-25Ø 4-20Ø	4- 25Ø 4- 20Ø	8-20Ø	4- 20Ø 4- 16Ø	8-16Ø	
12	E4	350 x 350	8-20Ø	8- 20Ø	4-20Ø 4-16Ø	4- 20Ø 4- 16Ø	4-20Ø 4- 16Ø	
13	E3	350 x 350	8-20Ø	8- 20Ø	4-20Ø 4-16Ø	4- 20Ø 4- 16Ø	4-20Ø 4- 16Ø	
14	E2	350 x 350	4-25Ø 4-20Ø	8- 20Ø	4-20Ø 4-16Ø	8- 16Ø	8-16Ø	

7.2 Design of beams

ETABS Concrete Frame Design

IS 456:2000 + IS 13920:2016 Beam Section Design (Summary)



Beam Element Details

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (mm)	LLRF	Type
Story2	B37	165	B 9" *14"	DL +YLL-EQXULS	177.8	3302	1	Ductile Frame

Section Properties

b (mm)	h (mm)	b_f (mm)	d_s (mm)	d_{ct} (mm)	d_{cb} (mm)
228.6	355.6	228.6	0	25	25

Material Properties

E_c (MPa)	f_{ck} (MPa)	Lt.Wt Factor (Unitless)	f_y (MPa)	f_{ys} (MPa)
27386.13	30	1	500	500

Design Code Parameters

γ_c	γ_s
1.5	1.15

Factored Forces and Moments

Factored M_{u3} kN-m	Factored T_u kN-m	Factored V_{u2} kN	Factored P_u kN
-89.605	8.3286	105.8438	0

Design Moments, M_{u3} & M_t

Factored Moment kN-m	Factored M_t kN-m	Positive Moment kN-m	Negative Moment kN-m
-89.605	12.5201	0	-102.1252

Design Moment and Flexural Reinforcement for Moment, M_{u3} & T_u

	Design -Moment kN-m	Design +Moment kN-m	-Moment Rebar mm²	+Moment Rebar mm²	Minimum Rebar mm²	Required Rebar mm²
Top (+2 Axis)	-102.1252		876	0	876	219
Bottom (-2 Axis)		0	438	0	20	438

Shear Force and Reinforcement for Shear, V_{u2} & T_u

Shear V_e kN	Shear V_c kN	Shear V_s kN	Shear V_p kN	Rebar A_{sv} /s mm²/m
109.78	0	168.073	83.7902	1408.79

Torsion Force and Torsion Reinforcement for Torsion, T_u & V_{u2}

T_u kN-m	V_u kN	Core b₁ mm	Core d₁ mm	Rebar A_{svt} /s mm²/m
8.3286	105.8438	198.6	325.6	717.23

Table 2: Beam reinforcement details

Beam Reinforcement Detail							
S.N.	Type of beam	Size of beam	Direction	Reinforcement			
				At support		At center	
				Top	Bottom	Top	Bottom
1	Tie beam	230*350	X-dir	3-16 ϕ Th	3-16 ϕ Th	2-16 ϕ Th	3-16 ϕ Th
			Y-dir	3-16 ϕ Th	3-16 ϕ Th	2-16 ϕ Th	3-16 ϕ Th
2	Ground Floor	230*350	X-dir	2-16 ϕ Th+1-12 ϕ Ex) +2-16 ϕ (Ex)	3-16 ϕ Th	2-16 ϕ Th	3-16 ϕ Th
			Y-dir	2-16 ϕ Th+1-12 ϕ Ex) +2-16 ϕ (Ex)	3-16 ϕ Th	2-16 ϕ Th	3-16 ϕ Th
3	Main beam First Floor, Second, Floor, Third Floor	230*350	X-dir	2-16 ϕ Th+1-12 ϕ Ex) +2-16 ϕ (Ex)	3-16 ϕ Th	2-16 ϕ Th	3-16 ϕ Th
			Y-dir	2-16 ϕ Th+1-12 ϕ Ex) +2-16 ϕ (Ex)	3-16 ϕ Th	2-16 ϕ Th	3-16 ϕ Th
4	Main beam Top Floor	230*350	X-dir	2-12 ϕ Th+1-16 ϕ (Ex)	3-12 ϕ Th	2-12 ϕ Th	3-12 ϕ Th
			Y-dir	2-12 ϕ Th+1-16 ϕ (Ex)	3-12 ϕ Th	2-12 ϕ Th	3-12 ϕ Th

7.3 Design of Slab

Design of Two way Slab by Limit State Method					
(As per IS 456:2000)					
1	Data				
	Longer span of slab, ly	4.521	m		
	Shorter span of slab, lx	3.454	m		
	Materials used				
	Concrete	M20	20	N/mm ²	
	Steel	Fe500	500	N/mm ²	
2	Effective span				
	Longer span of slab, ly	4.344	m		
	Shorter span of slab, lx	3.277	m		
	ly/lx	1.326			
	type of slab	<2 two way slab			
	Support Condition :				
	One Short Edge Discontinuous				
3	Calculation of loads				
	i) Dead load				
i	Slab thickness	125	mm		
	Self weight of concrete	25	KN/m ³		
	Dead load of slab = 25 * 0.125	3.125	KN/m ²		
ii	Screeding thickness	30	mm		
	Self weight of screeding	20.4	KN/m ³		
	Dead load of screeding = 20.4 * 0.03	0.612	KN/m ²		
iii	marble flooring thickness	19	mm		
	Self weight of marble	27	KN/m ³		
	Dead load of marble = 27 * 0.019	0.513	KN/m ²		
iv	Cement plaster thickness	12.5	mm		
	Self weight of plaster	20.4	KN/m ³		

	Dead load of plaster = $20.4 * 0.0125$	0.255	KN/m ²		
	Total dead load on slab	4.505	KN/m²		
	ii) Live load				
	For Residential Building	2	KN/m ²		
	Total design load (DL+LL)	6.505	KN/m²		
	Factored load = $1.5 * \text{design load}$	9.7575	KN/m ²		
4	Calculation of moments				
	As per code,	Shorter Direction		Longer Direction	
	-ve moment at support	Support, $\alpha x1$	0.052	Support, $\alpha y1$	0.037
	+ve moment at mid	mid, $\alpha x2$	0.040	mid, $\alpha y2$	0.028
	Along shorter span,				
	$M_x = \alpha x1 * w_u * l_x$	6.599	KNm		
	$M_y = \alpha y1 * w_u * l_x$	4.693	KNm		
	Along longer span,				
	$M_x = \alpha x2 * w_u * l_x$	5.012	KNm		
	$M_y = \alpha y2 * w_u * l_x$	3.552	KNm		
5	Depth Calculation				
	Minimum depth of slab is calculated considering maximum bending moment				
	Maxm bending moment	6.599			
	Depth required,				
	$d = \sqrt{\frac{M_u}{(0.36 * x_{u\max}/d) * (1 - 0.42 * x_{u\max}/d) * b * f_{ck}}}$				
	overall depth, $D = d + \phi/2 + \text{clear cover}$		mm		
	For safe design, adopting depth of slab	125	mm		
	Effective depth	96	mm		
6	Reinforcement Calculation	Main Bar	Dist. Bar		
	provide the reinforcement of dia	10	8	mm	
S. No.	Description	Shorter Direction		Longer Direction	
		Support	mid	Support	mid
1	αx & αy	0.052	0.040	0.037	0.028
2	BM	6.60	5.01	4.69	3.55
3	Ast required	149.81	112.75	105.40	79.25
4	Ast min	150.00	150.0	150.00	150.00

			0		
5	Spacing reqd	523.81	523.81	335.24	335.24
6	Spacing prov.	150.00	150.00	150.00	150.00
7	Ast provided	523.81	523.81	335.24	335.24
8	% of steel provided	0.42	0.42	0.27	0.27
7	Check				
7.1	Minimum reinforcement				
	Min Ast = 0.12% bD	150	mm ²		
7.2	Check for deflection				
	From deflection control criteria				
	$l_x/d < (l/d)_{basic} \cdot \alpha \beta \gamma$	where			
	l_x/d	31.2085714			
	α = modification factor for tension reinforcement				fig 4
	β = modification factor for compression reinf.	1			fig 5
	γ = reduction factor for ratios of span to eff. depth in flanged beam	1			fig 6
	$(l/d)_{basic} =$	23			of IS 456
	For α ,				
	$f_s = 0.58 \cdot f_y \cdot (A_{st \text{ required}} / A_{st \text{ provided}})$	83.045	N/mm ²		
	% of tensional reinforcement	0.419			
	α	2			
	now, $(l/d)_{basic} \cdot \alpha \beta \gamma$	46		OK Safe	
	Since, $l_x/d < (l/d)_{basic} \cdot \alpha \beta \gamma$	SAFE			
7.3	Check for Shear				
	Shear is checked along short span.				
	Maximum shear force, $V_{umax} = w \cdot l_x / 2$	19.354	KN		
	Normal shear stress, $\tau_v = V / bd$	0.184	N/mm ²		
	% of steel provided = $100 \cdot A_{st \text{ provided}} / bD$	0.249			
	Shear capacity of section without reinforcement				
	From table 19,				
	τ_c	0.360	N/mm ²		

	Shear strength, $\tau'c = k \cdot \tau_c$				
	For slab thickness 125mm, k	1.3			
	Shear strength, $\tau'c$	0.467	N/mm ²		
	Since, $\tau'c > \tau_v$	OK Safe			
8	Results				
	Overall depth of slab =	125	mm		
	Main bar along short span at support	10mm ϕ @ 150mm c/c			
	Main bar along short span at mid	10mm ϕ @ 150mm c/c			
	Distribution Bar along long span at support	8mm ϕ @ 150mm c/c			
	Distribution bar along long span at mid	8mm ϕ @ 150mm c/c			

7.4 Design of Staircase

Design of staircase

Unit weight of RCC=	25.0	kN/sq.m
Available width of Staircase=	1.3	m
Floor to Floor Height =	3.2	m
Height of Riser=	0.2	m
Width of tread=	0.3	
Number of Riser=	21.0	
Number of Tread required	20.0	

Design of Staircase AB

Effective span of flight=	Centre to centre distance of Support
=	2.3 m
Thickness of waist slab=	(1/20) of span(approx.)
=	78.3 mm

let us take

d=	100.0 mm
D=	125.0 mm

LOADS

i. weight of waist slab in plan (per m width of flight)

$$D \cdot \sqrt{1 + \left(\frac{R}{T}\right)^2} \cdot \text{unit weight} = 3.6 \text{ kN/m}$$

ii. Weight of steps (per m width of flight)= $\frac{(25RT)}{2T}$

$$1.9 \text{ kN/m}$$

Total dead load=	5.5 kN/m
Live load=	3.0 kN/m
Total DL + LL=	8.5 kN/m
Factored load, w_u =	12.8 kN/m

For landing,

DL=	3.1 kN/m
LL=	3.0 kN/m
Total, DL+LL=	6.1 kN/m
Factored load=	9.2 kN/m

Design Moment

Reaction at supports, R_A =	17.8 kN
Reaction at supports, R_B =	17.3 kN
BM at mid span, M_u =	17.9 kNm

Max. BM allowed for a singly reinforced

section with Fe500=

$$0.138 * f_{ck} * b * d * d$$

$$= 27.6 \text{ kNm} > 17.9 \text{ kNm}$$

Hence, section can be designed as singly reinforced

Area of Reinforcement

$$M_u = 0.87 * f_y * A_{st} * d * (1 - (A_{st} / b * d) * (f_y / f_{ck}))$$

Computing

$$A_{st} = 355.4 \text{ sq.mm}$$

$$\text{Using 10mm bars, } A_{steel} = 78.5 \text{ Sq.mm}$$

$$\text{Spacing} = A_{steel} / A_{st}$$

$$221.0 \text{ mm}$$

Considering spacing = 150mm

Area provided, $A_{prov} =$

$$523.6 \text{ sq.mm}$$

Provide 10mm dia. Bars @

$$150.0 \text{ c/c}$$

$$\text{Distribution Steel} = 0.12\% \text{ of area}$$

$$= 150.0 \text{ Sq.mm}$$

$$\text{using 10mm dia. Bars, } A_{steel} = 78.5 \text{ sq.mm}$$

$$\text{Spacing of 10 mm dia. Bars} = 523.6 \text{ mm}$$

$$\text{Providing spacing} = 200 \text{ mm}$$

$$A_{st} \text{ provided} = 392.7$$

provide 8mm dia. Bars @ 200mm c/c

Development length

$$L_d = (\text{Dia. Of steel} * 0.87 * f_y) / (4 * \text{shear strength})$$

$$546.9 \text{ mm}$$

Providing 600mm length of bars at points where L_d is required

Design of Staricase BB'

$$\text{Effective span of flight} = \text{Centre to centre distance of Support}$$

$$= 3.6 \text{ m}$$

$$\text{Thickness of waist slab} = (1/20) \text{ of span (approx.)}$$

$$= 124.3 \text{ mm}$$

let us take

$$d = 125.0 \text{ mm}$$

$$D = 150.0 \text{ mm}$$

LOADS

i. weight of waist slab in plan (per m width of

$$\text{flight}) D \cdot \sqrt{1 + \left(\frac{R}{T}\right)^2} \cdot \text{unit weight} \\ = 4.4 \text{ kN/m}$$

$$\text{ii. Weight of steps (per m width of flight)} = \frac{(25RT)}{2T} \\ 1.9 \text{ kN/m}$$

$$\begin{aligned} \text{Total dead load} &= 6.2 \text{ kN/m} \\ \text{Live load} &= 3.0 \text{ kN/m} \\ \text{Total DL + LL} &= 9.2 \text{ kN/m} \\ \text{Factored load, } w_u &= 13.8 \text{ kN/m} \end{aligned}$$

For landing,

$$\begin{aligned} \text{DL} &= 3.8 \text{ kN/m} \\ \text{LL} &= 3.0 \text{ kN/m} \\ \text{Total, DL+LL} &= 6.8 \text{ kN/m} \\ \text{Factored load} &= 10.1 \text{ kN/m} \end{aligned}$$

Design Moment

$$\begin{aligned} \text{Reaction at supports, } R_B &= 31.8 \text{ kN} \\ \text{Reaction at supports, } R_B' &= 34.1 \text{ kN} \\ \text{BM at mid span, } M_u &= 32.3 \text{ kNm} \end{aligned}$$

Max. BM allowed for a singly reinforced section with Fe500=

$$0.138 \cdot f_{ck} \cdot b \cdot d^2 \\ = 43.1 \text{ kNm} > 32.3 \text{ kNm}$$

Hence, section can be designed as singly reinforced

Area of Reinforcement

$$\begin{aligned} M_u &= 0.87 \cdot f_y \cdot A_{st} \cdot d \cdot \left(1 - \left(\frac{A_{st}}{b \cdot d}\right) \cdot \left(\frac{f_y}{f_{ck}}\right)\right) \\ \text{Computing} \\ A_{st} &= 426.4 \text{ sq.mm} \end{aligned}$$

$$\text{Using 10mm bars, } A_{steel} = 78.5 \text{ Sq.mm}$$

$$\begin{aligned} \text{Spacing} &= \frac{A_{steel}}{A_{st}} \\ &= 184.2 \text{ mm} \end{aligned}$$

Considering spacing = 115mm

$$\text{Area provided, } A_{prov} = 683.0 \text{ sq.mm}$$

Provide 8mm dia. Bars @

115.0 c/c

78.57143

Distribution Steel= 0.12% of area

180.0 Sq.mm

using 8mm dia. Bars, Asteel=

78.5 sq.mm

Spacing of 8 mm dia. Bars=

436.3 mm

Providing spacing=200mm

Ast provided=

392.7

provide 10 mm dia. Bars @ 200mm c/c

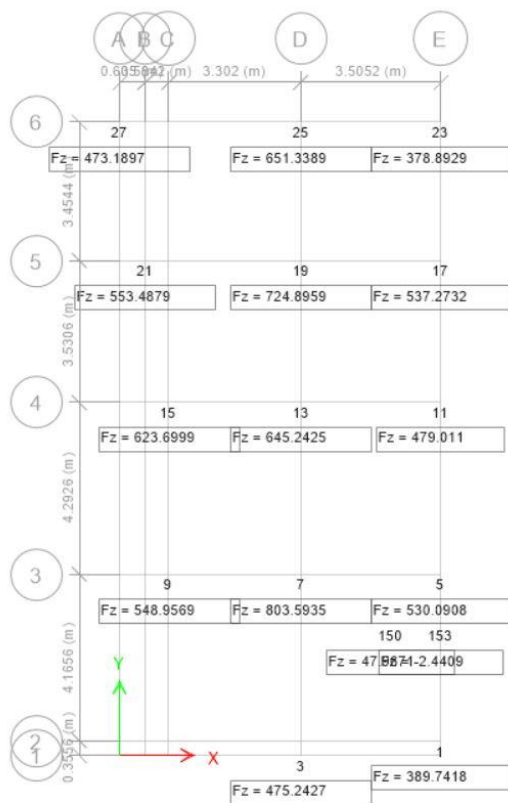
Development length

$L_d = (\text{Dia. Of steel} \times 0.87 \times f_y) / (4 \times \text{shear strength})$

546.9 mm

Providing 1000mm length of bars at points where L_d is required

7.5 Design of Footing



a. Isolated, Strap and Combined Footing is designed

b. Soil Subgrade Modulus

Factor of safety : 3 Soil Capacity : 120 KN/m²

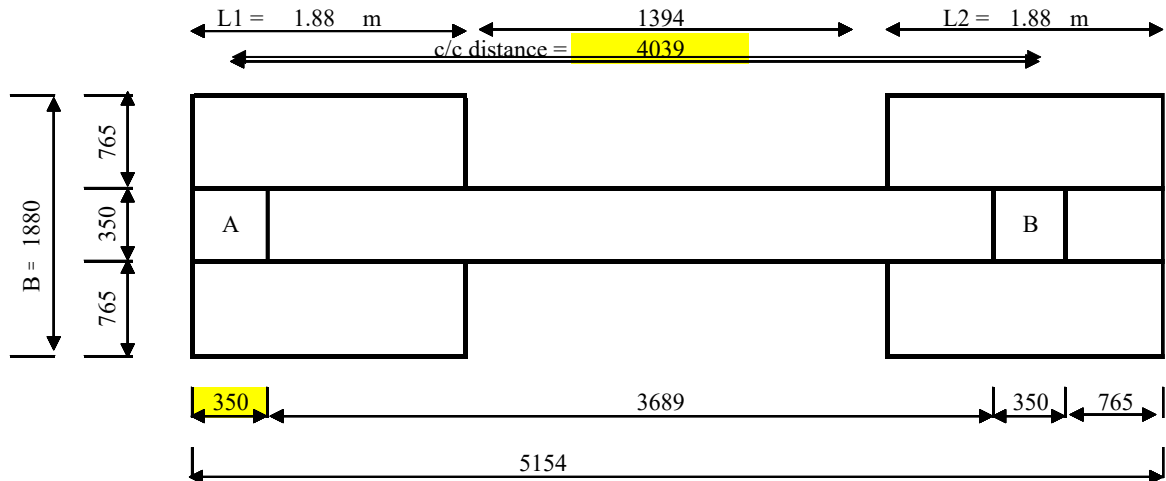
Allowable Settlement : 25 mm

Soil Spring Modulus = 3 x 150 x (1000/20)

= 225000 KN/m³

Preliminary slab thickness = 457.2 mm (18")

Design of Strap Beam (A3-E3)



PA	=	378.89	KN
PB	=	473.19	KN
SAFE BEARING CAPACITY, SBC	=	120	KN/m ²
Areqd, A	=	7.811	m ²
i.e., BL1+BL2	=	7.811	
cg of Forces from B, X	=	1.796	m
Let, B	=	1.88	m
		6.166	

$$A1(4.039+0.175-L1/2) = (A1+A2)X$$

$$BL1(4.214-L1/2) = (BL1+BL2)X$$

$$X = BL1(4.214-L1/2) / (BL1+BL2)$$

$$X = BL1(4.214-L1/2) / A$$

$$2*A*X/B = 8.428*L1-L1^2$$

$$L1^2 - 8.428*L1 + 2*A*X/B = 0$$

a	=	1
b	=	-8.43
c	=	14.92

L1	=	2.530	say, L1 =	1.88	m	6.1664
L2	=	1.624	say, L2 =	1.88	m	6.1664

T SOIL PRESSURE UNDER THE FOOTING	=	180.81	KN/m ²
ASSUMED DEPTH OF PAD	=	350	mm
ASSUMED DEPTH OF STRAP BEAM	=	600	mm
ASSUMED WIDTH OF STRAP BEAM	=	400	mm
CLEAR COVER TO MAIN REINF	=	50	mm
GRADE OF CONCRETE	=	20	N/mm ²
GRADE OF REINF	=	500	N/mm ²
DIA OF STRAP BEAM MAIN REINF	=	20	mm
DIA OF STRAP BEAM SHEAR REINF	=	8	mm
DIA OF PAD REINF	=	12	mm
EFFECTIVE DEPTH OF PAD	=	294	mm
EFFECTIVE DEPTH OF STRAP BEAM	=	540	mm

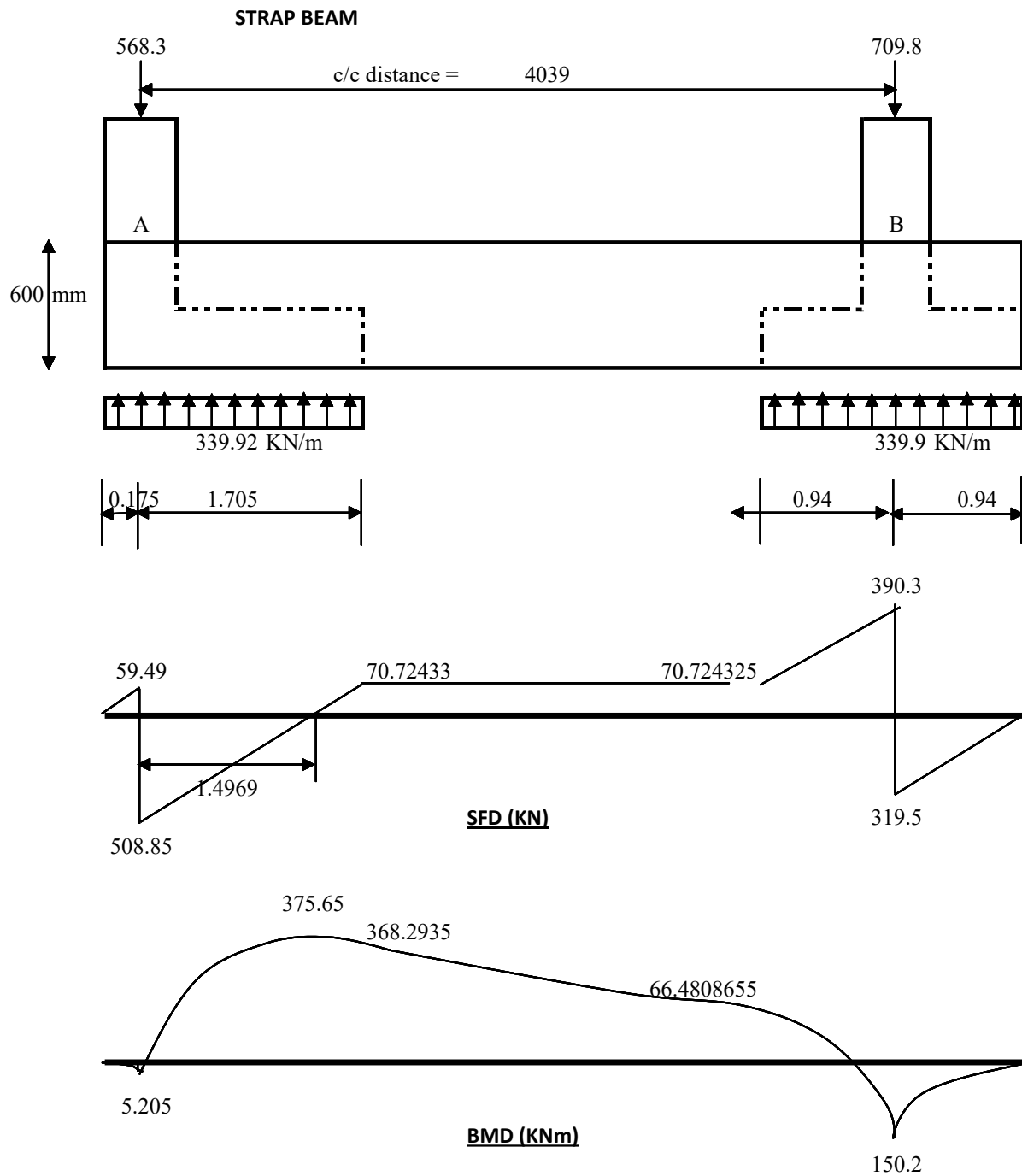
BENDING MOMENT AT BASE PAD

AT EXTERIOR FOOTING A, Mu	=	99.47 KN-m	
dreqd	=	138.45 mm	
	Mu = 0.87fy Ast(d-fy Ast/fck B)		
a =	$0.87fy^2/fckB$	=	5.78
b =	$-0.87fyd$	=	-127890
c =	Mu	=	99465957.75
Ast,reqd	=	807.22 mm ²	
Ast,min	=	789.60 mm ²	ok
Ast	=	807.22 mm ²	
Spacing,required	=	263.40 mm	
Spacing,provided	=	150.00 mm	ok

AT EXTERIOR FOOTING B, Mu	=	99.47 KN-m	
dreqd	=	138.45 mm	
	Mu = 0.87fy Ast(d-fy Ast/fck B)		
a =	$0.87fy^2/fckB$	=	5.78
b =	$-0.87fyd$	=	-127890
c =	Mu	=	99465957.75
Ast,reqd	=	807.22 mm ²	
Ast,min	=	789.60 mm ²	ok
Ast	=	807.22 mm ²	
Spacing,required	=	263.40 mm	
Spacing,provided	=	150.00 mm	ok

PAD SHEAR (ONE WAY SHEAR)

EFFECTIVE DEPTH, d	=	294 mm	
AT EXTERIOR FOOTING A, Vu	=	160.10 KN	
NOMINAL SHEAR STRESS, τ_v	=	0.290 N/mm ²	
ks	=	1	
$k\tau_c$	=	1.118 N/mm ²	ok
AT EXTERIOR FOOTING B, Vu	=	160.10 KN	
NOMINAL SHEAR STRESS, τ_v	=	0.290 N/mm ²	
ks	=	1	
$k\tau_c$	=	1.118 N/mm ²	ok



DESIGN OF STRAP BEAM FOR BM

FOR TOP REINFORCEMENT, M_u	=	368.29 KN-m	
d_{reqd}	=	588.34 mm	
$d_{provided}$	=	540.00 mm	FA
$M_u = 0.87f_y A_{st}(d - f_y A_{st}/f_{ck} B)$			
$a = 0.87f_y^2/f_{ck} B$	=	27.19	
$b = -0.87f_y d$	=	-234900	
$c = Mu$	=	368293458.8	
$A_{st, reqd}$	=	2058.15 mm ²	
$A_{st, min}$	=	288.00 mm ²	ok
A_{st}	=	2058.15 mm ²	
NO OF TOP BARS REQUIRED	=	6.55 mm	
NO OF TOP BARS PROVIDED	=	5.00 mm	FA

FOR BOTTOM REINFORCEMENT, M_u	=	150.18 KN-m	
d_{reqd}	=	368.82 mm	ok
$d_{provided}$	=	540.00 mm	ok
$M_u = 0.87f_y A_{st}(d - f_y A_{st}/f_{ck} B)$			
$a = 0.87f_y^2/f_{ck} B$	=	27.19	
$b = -0.87f_y d$	=	-234900	
$c = Mu$	=	150178342.1	
$A_{st, reqd}$	=	695.28 mm ²	
$A_{st, min}$	=	288.00 mm ²	ok
A_{st}	=	695.28 mm ²	
NO OF TOP BARS REQUIRED	=	2.21 mm	
NO OF TOP BARS PROVIDED	=	3.00 mm	ok

DESIGN OF STRAP BEAM FOR SHEAR

d FROM COLUMN FACE B TOWARDS A	=	147.21 KN	
d FROM COLUMN FACE A TOWARDS B	=	265.80 KN	
NOMINAL SHEAR STRESS, τ_v	=	1.231 N/mm ²	
%p	=	0.727 %	
SHEAR STRENGTH OF CONCRETE, τ_c	=	0.555 N/mm ²	
V_{us}	=	146.02 KN	
SPACING OF STIRRUPS REQUIRED	=	161.72 mm	
SPACING OF STIRRUPS PROVIDED	=	125 mm	ok

SUMMARY OF FOUNDATION

THICKNESS OF PAD	=	350	mm
WIDTH OF EXTERIOR ACROSS STRAP	=	1.88	m
LENGTH OF EXTERIOR ALONG STRAP	=	1.88	m
WIDTH OF INTERIOR ACROSS STRAP	=	1.88	m
LENGTH OF INTERIOR ALONG STRAP	=	1.88	m
PAD REBAR	=	TMT-12 @150c/c	
DEPTH OF STRAP BEAM	=	600	mm
WIDTH OF STRAP BEAM	=	400	mm
TOP BAR OF STRAP BEAM	=	5 Nos TMT-20	
BOTTOM BAR OF STRAP BEAM	=	3 Nos TMT-20	
STIRRUPS OF STRAP BEAM	=	TMT-8 @125c/c	
GRADE OF CONCRETE	=	M20	
CLEAR COVER TO REINF	=	50	mm

Design of Isolated Footing F4

Foundation Type : F4

Required Data

size of column=

l = 0.35 m

b = 0.35 m

Bearing Capacity of Soil= 120 KN/m²

Strength of Steel (fy)= 500 N/mm²

Strength of Concrete (fck)= 20 N/mm² (M 20)

Factor Load= 803.60

Axial Load = 535.73 KN

Approximate area of footing= 4.46 m²

Weight of footing including ea 133.93

Axial Load = 535.73 KN

Total weight on soil= 669.66 KN

Actual Area= 5.58 m²

Size of Square Footing = 2.362 7.7484

Adopted size=

L = 2.44 m 8.00

B = 2.44 m

Actual area of footing = 5.9536 m²

Net Pressure acting upwads = 134.98 KN/m²

B. M. at the face of column about an axis = 179.83 KN-m

The effective depth required is given by

$$BM = 0.138 fck * bd^2$$

d= 163.41 mm

Adopt d = 2 to 3 Times of calculated value of d for Shear considerations.

Adop. d= 375 mm

D= 425 mm

Check for one-way Shear action

The critical section is taken at distance d away from the face of column

$$\text{Shear force } V_u = 220.66 \text{ KN}$$

$$\text{Nominal shear stress } (T_v) = V_u / b * d = 0.24 \text{ N/mm}^2$$

$$\text{Allowable Shear Stress of Concrete for } \phi\% \text{ of Steel } < (M 20) = 0.28$$

$$\text{For } 0.165' \quad T_c = 0.292 \text{ N/sq.mm (From IS 456: 2000, table 19)}$$

Hence, **OK**

T_c should be equal to or greater than T_v

Check for two-way Shear action

The critical section is taken at a distance of $0.5d$ away from the face of column ,

$$\text{Shear force } V_u = 732.65 \text{ KN}$$

$$\text{Nominal Shear Stress } (T_v) = V_u / b' d \text{ (Where } b' \text{ is the periphery of critical section)}$$

$$\text{and } b' = 2900 \text{ mm}$$

$$\text{So, Nominal Shear Stress } (T_v) : 0.67 \text{ N/mm}^2$$

Shear Strength of concrete is

$$T_c = 1.12 \text{ N/sq.mm}$$

$$T'_c = K_s * T_c \text{ Where } K_s = (0.5 + B_c)$$

B_c = length of shorter side of column / length of longer side of column

$$B_c = 1$$

$$K_s = 1.5 \text{ should not be greater than } 1$$

$$\text{Take } (K_s) = 1$$

$$T'_c = 1.12 \text{ N/mm}^2$$

Hence, **OK** T'_c should be greater than T_v

Area of Steel Calculation,

Area of Steel is given by formula, $B_M = 0.87 * f_y * A_t * \{d - f_y * A_t / (f_{ck} * b)\}$

Then,

$$A_{st} = 1137.75 \text{ sq.mm} > A_{st \text{ min}} = 0.12\% \text{ of } b * d$$

$$A_{st \text{ per m}} = 466.29 \quad 450 \text{ mm}^2$$

$$\text{Req. Area of steel} = 466.29 \text{ mm}^2$$

$$\% \text{ of steel} = 0.124$$

Steel required as **12** mm bar @ 242.55 mm c/c distance.

Provide Steel as = 12 mm bar @ **150** mm c/c distance.

$$\text{Provided steel} = 753.98 \text{ mm}^2$$

Hence Ok.

Check for one-way Shear action

The critical section is taken at distance d away from the face of column

Shear force $V_u = 234.48$ KN

Nominal shear stress (T_v) = $V_u/b*d = 0.26$ N/mm²

Allowable Shear Stress of Concrete for pt% of Steel τ_c (M 20) = 0.28

For 0.165' $\tau_c = 0.292$ N/sq.mm (From IS 456: 2000, table 19)

Hence, **OK**

τ_c should be equal to or greater than T_v

Check for two-way Shear action

The critical section is taken at a distance of $0.5d$ away from the face of column ,

Shear force $V_u = 778.53$ KN

Nominal Shear Stress(T_v) = $V_u/b'd$ (Where b' is the periphery of critical section)

and $b' = 2900$ mm

So, Nominal Shear Stress(T_v) : 0.72 N/mm²

Shear Strength of concrete is

$\tau_c = 1.12$ N/sq.mm

$\tau_c = K_s * \tau_c$ Where $K_s = (0.5 + B_c)$

B_c = length of shorter side of column / length of longer side of column

$B_c = 1$

$K_s = 1.5$ should not be greater than 1

Take(K_s) = 1

$\tau_c = 1.12$ N/mm²

Hence, **OK** τ_c should be greater than T_v

Area of Steel Calculation,

Area of Steel is given by formula, $M = 0.87 * f_y * A_t * \{d - f_y * A_t / (f_{ck} * b)\}$

Then,

$A_{st} = 1211.51$ sq.mm $> A_{st \text{ min}} = 0.12\%$ of $b * d$

$A_{st \text{ per m}} = 496.52$ 450 mm²

Req. Area of steel = 496.52 mm²

% of steel = 0.132

Steel required as **12** mm bar @ 227.78 mm c/c distance.

Provide Steel as = 12 mm bar @ **150** mm c/c distance.

Provided steel = 753.98 mm²

Hence Ok.

Joint level	Unfactored load	Footing size(ft)	Depth of footing pad	Reinforcement bar
A2	445.98	7X7	450mm	12 mm 5" c/c Both ways
A3	348.72	6X6		
A4	294.25	6X6		
A5	193.88	5.5X5.5		
B1	435.23	7X7		
B2	9.98	6X6		
B3	569.28	8X8		
B4	486.48	8X8		
B5	343.82	6X6		
C1	229.40	6X6		
C2	362.78	7X7		
C3	247.81	6X6		
C4	304.65	6X6		
C5	233.59	6X6		

8.0 CONCLUSIONS AND RECOMMENDATION

Residential building with simple configuration has been fully designed using a computer program, ETABSv18. All the required design details of slab, beam, column and footing and staircase have been presented. Possible best-approaches have been adopted for most-economical design yet fulfilling all the requirements for important building. Due considerations have been given to the probable increase in load and strength as a partial factor of safety. Also, double tie-beams, on regular tie-beam at plinth level and one at top level of footing-pad is specified as the building is being constructed in high seismicity area. During construction, it is mandatory to follow IS13920, regarding provision of stirrups and ties, development lengths, splices and other detailing. Band at sill and lintel are provided to increase lateral strength of wall and to improve opening weakness on wall. Strict Control over quality of materials and workmanship is required for expected performance of building in future. Hence, following shall be considered during construction works to obtain expected results.

A. General:

- a. All works to be carried out in accordance with current best practice, Building Regulations, the project specification and relevant Nepal Building Code (NBC), Indian Standards and Codes of Practice. Materials and components to be appropriate for their intended use.
- b. The construction-works shall only be carried-out by trained mason with supervision of Engineer.
- c. During construction, the contractor shall be responsible for maintaining the structure in a stable condition and ensuring no part shall be damaged under construction activities.
- d. Workmanship and materials are to be in accordance with the relevant current Standards including all amendments and the local statutory authorities, except where varied by the contract document.
- e. All coarse aggregate used shall be crushed stone aggregate. The nominal size of coarse-aggregate for RC bands and splints shall not exceed 12.5mm.
- f. Clean sand, with minimum silt and free from clay and organic materials shall be used.
- g. Ordinary Portland cement conforming to IS 269:1976 shall be used for all cement works.
- h. At least 48 hours' notice shall be provided for all engineering inspections.

B. Structural:

1. Cast-In-Situ concrete/micro-concrete shall have minimum 28 days' compressive cube strength of 20N/mm² for all structural members unless otherwise stated.
2. The concrete compressive strength shall be measured on 150*150*150mm cube at 28 days, for various structural elements.
3. Reinforcing steel shall be, TMT rebar with ultimate strain not less than 14.5% can also be used.
4. Cover to main reinforcing steel be in accordance with IS 456:1978 & as specified in the structural drawings.

5. Clear Cover of Concrete shall not be less than that given below:
 - a. Concrete surface at soil = 50mm
 - b. Concrete on PCC, Bricks, STONE, etc. = 25mm
6. Unless otherwise specified, all horizontal & vertical construction joints shall be roughened.
7. A minimum of 48 hours' notice shall be given to the Engineer before applying plaster, concrete/microconcrete is poured, in order that the formwork and/or reinforcement may be inspected.
8. All R.C.C work shall be continuously cured for 14-days.
9. All cement plaster works shall be continuously cured for 7 days.
10. Any damage to surface during erection/construction is to be made good.

9.0 REFERENCES

- a. IS code 456-2000: Plain and Reinforced Concrete code of Practice
- b. IS code 1893 (part I)-2002: Criteria for earthquake Resistant Design of Structures
- c. IS code 875-1987: Code of Practice for Design Loads
 - Part 1 Dead Loads
 - Part 2 Imposed Loads
 - Part 5 Special loads and Load combination
- d. IS code 13920-1993: Ductile Detailing of RCC structures subjected to seismic force code of practice
- e. Nepal National Building Code (NBC 105): Seismic Design of Buildings in Nepal
- f. RCC structures by A.K. Jain
- g. RCC structures by S.N. Sinha
- h. Limit State Design by Vargees
- i. Structural Design of Multistoried building by U.H. Varyani
- j. IS code SP 16 Design guidelines for RCC Structures
- k. IS code SP 34 for detailing of RCC structures